Daily Growth at Risk: financial or real drivers?

The answer is not always the same

Helena Chuliá, Ignacio Garrón, Jorge Uribe

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Abstract

We propose daily Growth-at-Risk (GaR) for monitoring downside risks of the US economy, which uses financial and real high-frequency indicators. We show that the relative importance, in terms of forecasting power, of these indicators is time varying. Indeed, the optimal forecasting weights of our variables were clearly different during the Global Financial Crisis and the recent Covid-19 crisis, which reflects the dissimilar nature of the two crises. We introduce LASSO, Elastic Net, and adaptive sparse group LASSO into the family of mixed data sampling models used to estimate GaR and show that these outperform past candidates explored in the literature. Moreover, equity market volatility (VXO), credit spreads and the Aruoba-Diebold-Scotti business conditions index (ADS) are found to be relevant to nowcast economic activity, especially during crisis episodes. Overall, our results show that daily information for both real and financial variables is key for producing accurate point and tail risk nowcasts of economic activity.

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1. Introduction

There is a large body of recent studies that analyze the predicting power of financial conditions on real economic activity in times of crisis. One group emphasizes the significant role of financial indicators in forecasting low quantiles of the real Gross Domestic Product (GDP) growth rate (e.g., Giglio et al., 2016; Adrian et al., 2019), while other reports that after controlling for real variables, financial indicators have little to add to the mix (e.g., Reichlin et al., 2020; Plagborg-Møller et al., 2020). Indeed, some studies claim the opposite, i.e., that real variables (except for unemployment claims) have little to add after financial variables have been incorporated into the forecasting equation (see Carriero et al., 2022).

This disagreement occurs because forecasting real economic activity (or any fragment of the growth distribution for that matter) using financial variables is a unique and challenging problem due to two facts. First, financial and real variables are generally sampled at different frequencies, being the former of a considerably higher frequency than the latter. Second, quantifying how much financial variables add in terms of forecasting power for predicting economic activity, seems to be rather a causal question (in opposition to a predictive one), because it is related to the prolonged controversy in economics about the dichotomy between nominal and real variables, and how (and to what extent) the former influence the latter. Furthermore, this second issue highlights the tension that exists between pure "predictive" and pure "causal" tasks in social science and, specially, in economics (Athey, 2017). In short, this theoretical distinction is not crystal clear when it comes to macroeconomic studies, which are observational out of necessity, and in which forecasting can generally be improved by using domain knowledge, which is causal. Moreover, at the same time, forecasting exercises are generally expected to improve our understanding about the causal mechanisms in the economy. In short, we trust forecasts more than we can understand.

Due to this complex relationship and the multiplicity of aims that a researcher or policy maker may have for making a forecast, we recommend using an eclectic approach. In this approach, both financial and real variables should ideally be used to forecast economic crisis episodes, while letting the data show freely the relative importance of each set of variables on a time-varying basis. Following this approach, we make two important contributions to the field. First, we show that the informational content of financial and real economy daily indicators is different across time. Thus, in some circumstances, forecasting accuracy is due greatly to financial indicators, such as the equity market volatility index (VXO) or credit spreads; however, in other circumstances, real economic indicators, such as the Aruoba-Diebold-Scotti business conditions index (ADS) (Aruoba et al., 2009), are more useful for improving forecasts. Our results clearly show the time-varying importance of real and financial variables. We compute the optimal weights that our nowcasting Growth-at-Risk (GaR) models assign to the ADS index or to financial variables using forecast combination, and we show that in periods like the aftermath of the Global Financial Crisis (GFC), financial indicators play a far greater role than the ADS index, while the opposite is true for the recent Covid-19 crisis. This is in agreement with the general consensus in the macro-financial literature that highlights the financial nature of the GFC, when financial markets and intermediaries acted as amplifiers of shocks to the system (Isohätälä et al., 2016; Brunnermeier and Sannikov, 2016; Gertler and Gilchrist, 2018). However, past literature recognizes that the Covid-19 crisis was simply a product of the supply restrictions imposed to contain the pandemic, which were real and supply-side in nature, albeit with repercussions for the aggregate demand (Guerrieri et al., 2022). Note that understanding the mechanisms underlying a crisis episode is a purely causal task, which can assist the researcher to interpret the results of the forecasting exercise and also to improve the actual forecasting.

Second, we also contribute to the GaR literature, pioneered by Adrian et al. (2019) by using financial and real high frequency indicators. Unlike most of the literature that employs quarterly (e.g., Adrian et al., 2019; Brownlees and Souza et al., 2021) or weekly indicators (Carriero et al., 2022) to forecast tail risk to GDP growth, we estimate our models using daily right-hand-side variables. Thus, our results use more information in real-time than most of the extant literature. There are some exceptions in which daily data is also used (e.g., Ferrara et al., 2022; De Santis and Van der Veken, 2020); however, in these cases, real variables have been neglected and the number of financial indicators included is rather limited. Therefore, our results are also supported by richer cross-sectional information at the intended frequency than previous studies, which is reflected in the greater accuracy of our models compared to previous ones.

An important concern needs to be addressed when working with daily predictors and without excluding variables (financial or real) a priori from our set of predictors, as we do. The number of parameters to estimate increases quickly, compared to using lower frequencies and fewer variables. In this case, shrinkage, regularization and dimensionality reduction techniques, such as those provided by LASSO, the Elastic Net (EN) method, the adaptive sparse group LASSO or PCA, become essential. Thus, we introduce these methods into mixed data sampling (MIDAS)-Quantile models for estimating GaR. We use quantile regression for high dimensional spaces as proposed by Belloni and Chernozhukov (2011) and PCA to reduce the dimensions of our problem even further. In line with what has been emphasized by Lima et al. (2020) and Lima and Meng (2017), parameter-reduction techniques are crucial at high frequencies. LASSO-Quantile (LASSO-Q) tends to outperform other alternatives explored in past literature (i.e., the traditional quantile MIDAS, where the vector of high-frequency terms takes an arbitrary form, either estimated by frequentist (Ghysels et al., 2016) or Bayesian methods (Mogliani and Simoni, 2021; Ferrara et al., 2022), both, in and out-of-sample. In addition, in line with Stock and Watson

(2004), Andreou et al. (2013), and Ferrara et al. (2022), we show that combined forecasts using all indicators are more accurate, especially out-of-sample.

We validate our conclusions using a battery of statistics that come from the different fields of forecasting and quantitative risk-management. Rather than relying on a single statistic coming from the forecasting literature which, for instance, may not take into account that when the Value at Risk (VaR) of a series is estimated (low or high conditional quantile), two properties need to be achieved: Unconditional coverage and independence (Christoffersen, 1998). This is important because, as shown by Brownlees and Souza (2021) in a multicountry setting, original indicators of Growth at Risk frequently do not pass basic tests designed in finance to measure the precision of VaR estimates. This, in turn, may call into question the usefulness of the entire enterprise.

We show that this is not the case for our indicators. In fact, on the vast majority of occasions, daily financial information together with daily information on real activity are very useful for anticipating bad scenarios for GDP growth. In addition, our GaR statistics are adequate and meet the performance expected from them.

The rest of this document is organized as follows. Section 2 presents our data and section 3 our methodology. Section 4 presents our main results, while section 5 concludes.

2. Data

In our exercise, we are interested in nowcasting the conditional tail of the distribution of real GDP growth (or GNP for some sample period[1]) on a pseudo-real-time basis. To do this, we use

[1] In December 1991, the Bureau of Economic Analysis switched from reporting GNP to reporting GDP as the output measure. Also, in January 1996, they switched from reporting fixed-weight aggregation methods to chain-weight methods in GDP.

the quarterly real-time data set collected by the Federal Reserve Bank of Philadelphia (FRBP) from 1986Q1 to 2020Q4. Specifically, this dataset gathers the advance estimate for the previous quarter, which is released near the end of the month of the current quarter[2].

Regarding the daily high-frequency variables, we include 12 daily predictors to compute our GaR forecasts (11 financial variables and 1 real variable). From this set, eight series are the same as in Pettenuzzo et al. (2016): i) the ADS daily business cycle index of Aruoba et al. (2009), which comes from a dynamic factor model at daily frequency; ii) the interest rate spread between the 10-year government bond rate and the federal funds rate (ISPREAD); iii) the change in the effective federal funds rate (EEFR); iv) the BAA-AAA corporate bond yield credit spread (CSPREAD); v) the excess return on the market (RET); vi) the returns on the portfolio of small minus big stocks (SMB); vii) the returns on the portfolio of high minus low book-to-market ratio stocks (HML); and viii) the returns on a winner minus loser momentum spread portfolio (MOM). In addition to this group, we add four financial indicators: the equity market volatility index (VXO), which has been used as a risk indicator[3]; the spread between the yield of 10-year constant maturity Treasury bonds and 3-month Treasury bills (TERM), as a predictor of US recessions[4]; the spread between 3-month Libor based on US dollars and 3-month Treasury bills spread (TED), as a proxy of credit risk[5]; and the Composite Indicator of Systemic Stress (CISS) for the US, which is a systemic risk measure based on 15 raw market indicators, following a computation analogous to the CISS for the Euro Area (Holló et al., 2012). The latter variable is

[2] Faust et al. (2013) use this dataset to forecast real-time measures of economic activity using Bayesian Model Averaging with a large number of real and financial indicators.
[3] Rey (2015) shows that this indicator comoves with global capital flows, global credit growth, and global asset prices. Longstaff et al. (2011) also document that the price of sovereign risk is strongly correlated with VXO.
[4] Estrella and Mishkin (1998) and subsequent literature has shown the forecasting power of the term spread for recessions.
[5] Gunay (2020) show that the TED spread is superior to credit default swap indexes as an early warning indicator for the credit market.

used as a benchmark indicator in our models, as the standard GaR framework considers a composite financial condition index (Adrian et al., 2019; Figueres and Jarociński, 2020). Our data sample spans the period from 1986Q1 to 2020Q4 and is restricted by the availability of data for all indicators.

The ADS index is used in our nowcasting exercise with weekly vintages starting Nov 30, 2008. Although weekly vintages reduce uncertainty at the sample endpoints (Amburgey and McCracken, 2022), there is still uncertainty due to the estimation of the ADS index in a previous step. Maldonado and Ruiz (2021) emphasize the importance of accurately measuring this type of uncertainty in empirical applications. This means, for instance, that favourable economy conditions (better than average) can only be confidently asserted if both, the point estimate of the ADS index and its confidence intervals are positive. This uncertainty is inherent in our GaR models, which use the ADS index as an input variable and may produce overstated results (making estimates to look more precise of what they actually are). This limitation applies to all nowcasting exercises using an index estimated in a previous step (not only the ADS), and must be acknowledged. We include up to one year of daily lags of the high-frequency indicator in all our specifications. The detailed description of these indicators is shown in Table 1.

Table 1. Detail description of variables

Variable	Frequency	Sample	Lags	Description	Source
ADS	Daily	Jan. 1, 1986, to Dec. 31, 2020	1 year	ADS index weekly vintages collected in real-time from November 30, 2008.	Federal Reserve Bank of Philadelphia
ISPREAD	Daily	Jan. 1, 1986, to Dec. 31, 2020	1 year	Interest rate spread between the 10-year government bond rate and the federal fund rate.	Federal Reserve Bank of St. Louis
EEFR	Daily	Jan. 1, 1986, to Dec. 31, 2020	1 year	Effective Federal Funds Rate, first difference.	Federal Reserve Bank of St. Louis
CSPREAD	Daily	Jan. 1, 1986, to Dec. 31, 2020	1 year	BAA-AAA corporate bond yield credit spread.	Federal Reserve Bank of St. Louis
RET	Daily	Jan. 1, 1986, to Dec. 31, 2020	1 year	Excess return on the market, value-weight return of US stocks.	Fama and French (1993)
SMB	Daily	Jan. 1, 1986, to Dec. 31, 2020	1 year	The average return on the three small portfolios minus the average return on the three big portfolios.	Fama and French (1993)
HML	Daily	Jan. 1, 1986, to Dec. 31, 2020	1 year	The average return on the two value portfolios minus the average return on the two growth portfolios.	Fama and French (1993)
MOM	Daily	Jan. 1, 1986, to Dec. 31, 2020	1 year	The average return on the two high prior return portfolios minus the average return on the two low prior return portfolios.	Fama and French (1993)
VXO	Daily	Jan. 1, 1986, to Dec. 31, 2020	1 year	Option-based implied volatility measure of S&P100.	Federal Reserve Bank of St. Louis Federal Reserve Bank of St. Louis
TERM	Daily	Jan. 1, 1986, to Dec. 31, 2020	1 year	Spread between the yield of 10-year constant maturity Treasury bonds and of 3-month Treasury bills.	Federal Reserve Bank of St. Louis
TED	Daily	Jan. 1, 1986, to Dec. 31, 2020	1 year	Spread between 3-Month LIBOR based on US dollars and 3-month Treasury bills.	Federal Reserve Bank of St. Louis

CISS	Daily	Jan. 1, 1986, to Dec. 31, 2020	1 year	Daily systemic risk measure based on Holló et al. (2012).	European Central Bank
GDP growth	Quarterly	Q1, 1986, to Q4, 2020	1 quarter	Real GDP or Gross National Product, percent change from preceding period, quarterly, seasonally adjusted.	Federal Reserve Bank of Philadelphia

Source: Authors' own elaboration.

3. Methodology

To nowcast tail risks in GDP growth, we extend the formulation of Adrian et al. (2019) to account for daily high-frequency predictors. In this section we show concisely how we adapt the standard GaR to incorporate daily financial and real indicators, using mixed data sampling. To this end, we compare the performance of a traditional MIDAS (with Almon lag polynomials), Bayesian MIDAS, LASSO, EN, adaptive sparse group LASSO, and soft and hard thresholding methods. We also show how we implement combination forecasts, which have been shown to improve forecast accuracy. Finally, we present the tools used to evaluate tail risk forecasts.

3.1. Growth at Risk framework

As in the standard framework of quarterly GaR pioneered by the works of Giglio et al. (2016) and Adrian et al. (2019), we rely on quantile regressions (Koenker and Bassett, 1978). Specifically, we assess the combined effect of past GDP growth (y_{t-h}) and a given financial condition indicator (x_{t-h}), at quarter t and forecast horizon h, on the current output growth (y_t). At this point it is important to recall that even though x_{t-h} is observed daily, it is aggregated to quarterly frequency by taking its average.

The baseline quantile regression is given by:

$$y_t = \beta_0(\tau) + \beta_1(\tau)y_{t-h} + \beta_2(\tau)x_{t-h} + \epsilon_t(\tau)\,, \tag{1}$$

where $\beta(\tau) = (\beta_0(\tau), \beta_1(\tau), \beta_2(\tau))'$ denotes the vector of parameters corresponding to the τ-th quantile, and $\epsilon_t(\tau)$ is a random noise.

The parameters in Eq. (1) are estimated by minimizing the tick loss (TL) function:

$$TL_\tau = \frac{1}{T}\sum_{t=h+1}^{T}\left[\rho_\tau\left(y_t - Q_\tau(y_t|y_{t-h}, x_{t-h})\right)\right], \tag{2}$$

where $\rho_\tau(\epsilon_t) = (1-\tau)\mathbb{1}_{\{\epsilon_t<0\}}|\epsilon_t| + \tau\mathbb{1}_{\{\epsilon_t>0\}}|\epsilon_t|$, with $\mathbb{1}_{\{\epsilon_t<0\}}$ taking the value of 1 when the subscript is true and 0 otherwise. The mathematical formulation in Eq. (2) leads to the solution of a linear programming optimization problem that we have not included here. Its basic structure and the counterpart algorithm solution can be found in Koenker (2005).

The predicted value from Eq. (1) is the quantile of $y_{T|T-h}$, which is conditional on the information available up to $T-h$,

$$Q_\tau(y_T|y_{T-h}, x_{T-h}) = \beta_0(\tau) + \beta_1(\tau)y_{t-h} + \beta_2(\tau)x_{t-h}, \tag{3}$$

Koenker and Bassett (1978) further prove that the predicted value $Q_\tau(y_T|y_{T-h}, x_{T-h})$ is a consistent linear estimator of the conditional quantile function of y_t. In this setting, we are particularly interested in the $GaR(10\%)$ measure defined as the conditional 10% quantile forecast (see Figueres and Jarociński, 2020; Ferrara et al., 2022; Carriero et al. 2022), namely $Q_{\tau=10\%}(y_T|y_{T-h}, x_{T-h})$[6].

This last equation can be interpreted as the 10% quantile of GDP growth, which is conditional on the information set available up to $T-h$ for the predictors. On one hand, there is a vast literature that documents that financial conditions have strong predictive information for the lower quantiles of future GDP growth (see e.g., Adrian et al., 2019; Prasad et al., 2019; Brownlees

[6] Alternatively, Adrian et al. (2019) use the 5 percent quantile forecast as the measure of tail risk. However, due to a shorter sample compared to these works, we opt to use the 10 percent quantile.

and Souza, 2021; Figueres and Jarociński, 2020; Ferrara et al., 2022). However, on the other hand, Plagborg-Møller et al. (2020) and Reichlin et al. (2020) state that controlling for real factors is necessary to accurately measure the real-time effect of financial indicators on real activity. We acknowledge these two results in our framework by incorporating the ADS daily business cycle index of Aruoba et al. (2009), in addition to the financial indicators, as a high-frequency indicator of the real sector. We then follow a combination approach to produce a better point forecast and check the optimal weights of individual high frequency predictors, following previous literature in this field (see Stock and Watson, 2004; Andreou et al., 2013; Pettenuzzo et al., 2016; Ferrara et al., 2022).

3.2. Adapting the standard GaR approach to high-frequency indicators

The problem of the formulation in Eq. (1) is that the aggregation of the high-frequency indicator prevents the model from responding to daily shocks. Consequently, as in Ferrara et al. (2022), we adapt it to account for the daily information flow of the high-frequency indicator up to the latest available observation (minus h_d days), based on the following regression:

$$y_t = \beta_0(\tau) + \beta_1(\tau) y_{t-1} + \boldsymbol{X}_{t-h_d}^{D}{}' \, \boldsymbol{\phi}(\tau) + \epsilon_t(\tau) \,, \tag{4}$$

where $\boldsymbol{\phi}(\boldsymbol{\tau})$ is a $p \times 1$ vector of daily parameters and $\boldsymbol{X}_t^D = (x_t^0, x_t^1, \dots, x_t^{p-1})'$ is a $p \times 1$ vector of the high-frequency variable available on a daily basis, with $x_t^j, j = (0,1,2, \dots, p-1)$, which is updated d times between quarter t and $t-1$. In this setup, we consider that y_t is affected by up to one year ($q = 4$ quarters) of past daily shocks and past GDP growth, giving a total number of parameters (including the constant) approximately equal to $K = q * d + 2 = 4 * 60 + 2 = 242$, assuming a five-day working week $(d = 60\ days)$; that is, $\boldsymbol{X}_{t-h_d}^{D} =$

$(x_t^0, x_{t-\frac{1}{60}}^1, \dots, x_{t-\frac{239}{60}}^{239})'$, with $x_{t-h_d}^j$. Notice that in this formulation, the forecast horizon is expressed in high-frequency terms, that is, $h_d = (0,1/d, 2/d, \dots, (p-1)/d)$.

Our estimation window is wider than that of Ferrara et al. (2022), as they consider a lag window of 60 days for the high-frequency indicator. This enables the model to capture up to one year of daily information. In our case, the number of parameters K is relatively larger than the total number of observations T, so we are faced with a parameter proliferation problem, which invalidates the standard estimation procedure of the quantile regression. Thus, in what follows we discuss the four alternative methods used in our results section to estimate the above regression.

3.2.1. MIDAS-Q

The MIDAS-Quantile model (MIDAS-Q) offers an effective solution to incorporate high-frequency indicators into Eq. (4). This relies on a restriction of the form in which the distributed lags of the high frequency variable are included in the regression. Specifically, we introduce the high-frequency lagged vector $\boldsymbol{X}_{t-h_d}^D$ in a quantile regression for the low-frequency dependent variable y_t as follows:

$$y_t = \beta_0(\tau) + \beta_1(\tau) y_{t-1} + \sum_{j=0}^{p-1} b(j; \boldsymbol{\theta}(\tau)) L^{\frac{j}{d}} x_{t-h_d}^j + \epsilon_t(\tau)\,, \tag{5}$$

where $b(j; \boldsymbol{\theta}(\tau)) = \sum_{i=0}^{c} \theta_{i,j}(\tau) j^i$ is the Almon lag polynomial weighting function that depends on the vector of parameters $\boldsymbol{\theta}(\tau)$, where $j = (0,1,2, \dots, p-1)$, and the order of the Almon lag polynomial is given by c. While Ghysels et al. (2016) propose the Beta lag polynomial function for the quantile weighting function, we consider the Almon lag polynomial as in other

recent works (see Lima et al., 2020; Mogliani and Simoni, 2021; Ferrara et al., 2022). Under the so-called "direct method", Eq. (5) can be reparametrized as follows:

$$y_t = \beta_0(\tau) + \beta_1(\tau) y_{t-1} + \widetilde{\boldsymbol{X}}_{t-h_d}^{D}{}' \boldsymbol{\phi}(\tau) + \epsilon_t(\tau)\,, \tag{6}$$

where $\widetilde{\boldsymbol{X}}_{t-h_d}^{D} := \boldsymbol{Q} \times \boldsymbol{X}_{t-h_d}^{D}$ is a $(c+1) \times 1$ vector representing the transformed high-frequency predictor, $\boldsymbol{Q}$ is a $((c+1) \times p)$ weighting matrix with the $(i - th + 1)$ row element of $\boldsymbol{Q}$ equal to $\left(0^i, 1^i, 2^i, \ldots, (p-1)^i\right)$ for $i = 0, \ldots, c$. Following Ferrara et al. (2022), we set $c = 3$ (third degree Almon lag) and impose two end-point zero restrictions on the slope and the value of the lag polynomial $(r = 2)$, such as $b\left(p-1; \theta_j(\tau)\right) = 0$ and $\nabla_j b\left(j; \theta_j(\tau)\right)|_{j=p-1} = 0$, as in Mogliani and Simoni (2021). This causes the weighting structure to slowly reduce to zero. Consequently, the number of parameters of the high-frequency indicator to be estimated is reduced from $(c+1)$ to $(c+1-r)$ parameters.

3.2.2. BMIDAS-Q

The Bayesian version of the MIDAS-Quantile model (BMIDAS-Q), based on the Asymmetric Laplace Distribution (ALD) estimation approach pioneered by Yu and Moyeed (2001), offers a convenient alternative for estimating Eq. (6). Specially, this approach is considered by Ferrara et al. (2022) to nowcast GaR for the Eurozone using high-frequency financial indicators. Yu and Moyeed (2001) showed that the minimization problem of quantile regressions (see Eq. (2)) is equivalent to maximizing the likelihood function using the ALD for the error term $\epsilon_t(\tau)$. We use the Gibbs sampler implementation of Kozumi and Kobayashi (2011), alongside their mixture representation.

In this framework, the error term $\epsilon_t(\tau)$ in Eq. (6) can be represented as a location scale mixture of normal distributions in which the mixing distribution follows an exponential distribution (see Kozumi and Kobayashi, 2011).

This implies that Eq. (6) can be expressed as:

$$y_t = \beta_0(\tau) + \beta_1(\tau)y_{t-1} + \widetilde{\boldsymbol{X}}_{t-h_d}^{D}{}' \boldsymbol{\phi}(\tau) + \varphi_1(\tau)\, v_t + \varphi_2(\tau)\sqrt{\sigma v}_t u_t\,, \qquad (7)$$

where φ_1 and φ_2 are fixed parameter functions of the quantile τ, $v_t = \sigma z_t$ follows a standard exponential function, and u_t is a standard normal function. This leads to the following likelihood function (to simplify assume $\boldsymbol{X_t}$ contains all covariates):

$$f(y_t|\boldsymbol{X_t'}\boldsymbol{\phi}(\tau), v_t, \sigma) \propto \exp\left(-\sum_{t=1}^{T}\frac{\left(y_t-\boldsymbol{X_t'}\boldsymbol{\phi}(\tau)-\varphi_1(\tau)v_t\right)^2}{2\varphi_2(\tau)^2\sqrt{\sigma v_t}}\right)\prod_{t=1}^{T}\frac{1}{\sqrt{\sigma v}},$$

with posterior densities for $\boldsymbol{\phi}$, v and σ given by:

$$\boldsymbol{\phi}|X, v, \sigma, \tau \sim N(\tilde{\beta}, \tilde{V}\ ),$$

$$v|X, \boldsymbol{\phi}, \sigma, \tau \sim GiG\left(\frac{1}{2}, \frac{\left(y_t-\boldsymbol{X_t'}\boldsymbol{\phi}(\tau)\right)^2}{\sigma\varphi_2(\tau)^2}, \frac{2}{\sigma} + \frac{\varphi_1(\tau)^2}{\sigma\varphi_2(\tau)}\right),$$

$$\sigma|X, v, \boldsymbol{\phi}, \tau \sim \textit{Inv-Gamma (a, b)},$$

where

$$\tilde{\beta} = \tilde{V}\left(\sum_{t=1}^{T}\frac{\boldsymbol{X_t}(y_t-\varphi_1(\tau)v_t)}{\varphi_2(\tau)^2\sigma v_t}\right) \text{ and } \tilde{V}^{-1} = \sum_{t=1}^{T}\frac{\boldsymbol{X_t'}\boldsymbol{X_t}}{\varphi_2(\tau)^2\sigma v_t} + \tilde{V}_0^{-1},$$

and a and b are shape and scale parameters of the inverse gamma distribution, respectively. In our framework, as in Carriero et al. (2022), we are interested in using the posterior mean of the coefficient vector $\boldsymbol{\phi}(\tau)$ to produce point forecasts.

3.2.3. Soft thresholding: LASSO-Q and Elastic Net-Quantile (EN-Q)

One caveat of the restricted MIDAS approach presented above is the predetermined choice of the weighting function, which may lead to a lag structure for the high-frequency predictor that does not maximize forecast accuracy. Therefore, alternatively we propose estimating GaR by using either the LASSO or EN regularization methods for choosing a lag structure for the high-frequency predictors (Bai and Ng, 2008; Lima et al., 2020).

Accordingly, we select the lags of the high frequency variable, based on the LASSO-Q algorithm proposed by Belloni and Chernozhukov (2011). The model can be summarized as follows,

$$\min_{\beta,\phi} E\left[\rho_\tau(y_t - \beta_0(\tau) - \beta_1(\tau)y_{t-1} - \boldsymbol{X}^{\boldsymbol{D}}_{\boldsymbol{t-\mathrm{h}_d}}{}' \boldsymbol{\phi}(\boldsymbol{\tau}))\right] + \lambda_\tau \left[\frac{\sqrt{\tau(1-\tau)}}{T}\right] \sum_{i=0}^{p-1} |\phi_i(\tau)| \,, \quad (8)$$

where the optimization problem is the sum of the standard quantile minimization function (as in Eq. (2)) and a penalty function given by a scaled l_1-norm of the daily vector of parameters $\phi_i(\tau)$. The overall penalty is given by $\lambda_\tau[\sqrt{\tau(1-\tau)}/T]$, where T is the sample size. The optimal level of λ_τ (LASSO-Q penalization) is calculated as in Belloni and Chernozhukov (2011). The LASSO-Q penalty has the distinctive feature of making the coefficients of insignificant predictors exactly equal to zero, keeping only the informative predictors for the forecast.

The EN estimator of Zou and Hastie (2005) aims to solve two potential drawbacks of the original LASSO. First, if $K > T$, LASSO can select T variables at most. Second, if there is a group of variables with high pairwise correlation coefficients, LASSO tends to select only one variable from the group and does not care which one. Both, LASSO and EN shrink the estimates and perform model selection. However, while the LASSO penalty is convex, the EN penalty is strictly convex, which means that predictors must be grouped to have similar coefficients. The EN-Q objective function is given by:

$$\min_{\beta,\phi} E\,[\rho_\tau(y_t - \beta_0(\tau) - \beta_1(\tau)y_{t-1} - \boldsymbol{X^D}_{\boldsymbol{t-\mathrm{h}_d}}{}'\boldsymbol{\phi}(\tau))] +$$

$$\lambda_{1,\tau}\sum_{i=0}^{p-1}|\phi_i(\tau)| + \lambda_{2,\tau}\sum_{i=0}^{p-1}\phi_i(\tau)^2\,, \tag{9}$$

where $\boldsymbol{\lambda_{1,\tau}}$ and $\boldsymbol{\lambda_{2,\tau}}$ are two tuning parameters that satisfy $\frac{\lambda_{2,\tau}}{\lambda_{1,\tau}+\lambda_{2,\tau}} > 0$. This restriction implies that the EN-Q is strictly convex, so it forces high pairwise correlated predictors to have similar coefficients. As a result, EN-Q stretches the net that retains all the important predictors, even if they are highly correlated.

As shown by Zou and Hastie (2005), we can reformulate the EN objective function as a LASSO problem[7]. This has computational appealing properties, as we can use the Belloni and Chernozhukov (2011) algorithm to estimate the EN-Q model. To implement this idea lets define (for simplicity assume that $\boldsymbol{X_t}$ contains all covariates): $y_t^+(\tau) = \left(y_t\ O_p\right)'$and $\boldsymbol{X_t^+}(\tau) = \frac{1}{\sqrt{1+\lambda_{2,\tau}}}\left(\boldsymbol{X_t}\ \sqrt{\lambda_{2,\tau}}\boldsymbol{I_p}\right)'$, where O_p represents a $p\ \times\ 1$ vector of zeros and $\boldsymbol{I_p}$ is a $p \times p$ identity matrix.

Based on this new formulation, Eq. (9) can be re-expressed as follows,

$$\min_{\boldsymbol{\phi}^{++}} E\,[\rho_\tau(y_t^+ - \boldsymbol{X_t^+}'\boldsymbol{\phi}(\tau))] + \gamma_\tau\sum_{i=0}^{p-1}|\phi_i(\tau)|, \tag{10}$$

where $\gamma_\tau = \frac{\lambda_{1,\tau}}{\sqrt{1+\lambda_{2,\tau}}}$. Notice that now the sample size is equal to $T\ +\ p$, which enables EN to select all p high-frequency predictors in all cases. To remove the double shrinkage effect from LASSO, the EN-Q estimator is $\boldsymbol{\phi}^+(\tau) = (1+\lambda_{2.\tau})\boldsymbol{\phi}^{++}(\tau)$ (see Zou and Hastie, 2005). In our

[7] Bai and Ng (2008) and Lima et al. (2020) use this approach to produce conditional mean forecasts with different loss functions. The former apply the mean square error and the latter the tick loss function.

application, we only apply this correction if we use the EN-Q model to directly produce the conditional quantile nowcasts.

3.2.4. Soft and hard thresholding methods: LASSO-PCA-Q and EN-PCA-Q

We consider the approach of soft and hard threshold methods applied to forecasting with many predictors, in the same vein as Lima et al. (2020) and Bai and Ng (2008). To implement this, first, we estimate principal components from the non-zero coefficients selected by LASSO-Q or EN-Q. Then, from these selected variables, we estimate factors by principal components analysis and select the optimal number of factors using the eigen ratio (Ahn and Horenstein, 2013). Finally, we keep the factors with associated p-values lower than 0.01 (or the statistically most significant ones). To differentiate these models from those that only use the soft threshold to produce forecasts (i.e., LASSO-Q and EN-Q), we label these types of models as LASSO-PCA-Q and EN-PCA-Q, with the first step selected by LASSO-Q or EN-Q, respectively.

3.2.5. Adaptive sparse group LASSO (ASGL-Q)

We introduce a novel framework based on an additive sparse group LASSO-Quantile (ASGL-Q) regressions proposed by Mendez-Civieta et al. (2021)[8]. This technique is especially suited for high-dimensional problems were $(p >> T)$ as for dealing with multiple groups of high-frequency variables, with sparsity allowed within the high-frequency lagged vector. Also, it uses adaptive weights in the penalization scheme, in line with Zou (2006). For notation simplicity, we refer to $TL_{\tau}(\boldsymbol{\phi}(\boldsymbol{\tau}))$ as the TL function of the vector of parameter $\boldsymbol{\phi}(\boldsymbol{\tau})$ defined (see Eq. (2)).

The *ASGL-Q* objective function is given by:

[8] We thank an anonymous reviewer for suggesting this model.

$$\min_{\phi} E\left[TL_{\tau}(\boldsymbol{\phi}(\boldsymbol{\tau})) + \alpha\lambda \sum_{j=0}^{n-1} \widetilde{w}_j \left|\phi_j(\tau)\right| + (1-\alpha)\lambda \sum_{l=0}^{m-1} \sqrt{p_l}\tilde{v}_l \|\phi(\tau)^l\|_2\right], \quad (11)$$

where $\widetilde{w}_j$ is the weight for the $j-th$ parameter $\phi_j(\tau)$, $\tilde{v}_l$ is the weight for the $l-th$ group of parameters (or high-frequency variable) $\phi(\tau)^l$, and p_l is the size of the $l-th$ group. Overall, these weights assign a low weight to a relatively important high-frequency variable (or to a given lag), thus penalizing less. Notice that Eq. (11) is a linear combination of LASSO and group LASSO, given by λ and the tradeoff between the two of them, $\alpha \in [0,1]$. Specifically, a value close to 1 leads to additive LASSO and close to 0 to additive sparse group LASSO. Thus, this formulation provides solutions that are both, between and within groups. Also, as pointed out by Mendez-Civieta et al. (2021), this formulation defines a convex function which ensures that solution of the minimization process is a global minimum.

3.3. Forecast Combination

There is extensive literature that documents that forecast combinations provide a superior performance, as they use the information from all the underlying models rather than relying on one specific model (e.g., Stock and Watson, 2004; Andreou et al, 2013; Pettenuzzo et al. 2016; Ferrara et al., 2022). Indeed, selecting just one model can be misleading and highly inconvenient in the presence of misspecification (Hansen et al., 2011). While there are different methods for implementing forecast combinations, we rely on the discounted mean squared forecast error combinations approach (Stock and Watson, 2004; Andreou et al, 2013), using the TL as the objective function.

Combination weights are computed recursively on a daily basis in the following way:

$$w_{i,t-h_d} = \frac{\lambda_{i,t-h_d}^{-\kappa}}{\sum_i^N \lambda_{i,t-h_d}^{-\kappa}},$$

$$\lambda_{i,t-h_d} = \sum_{s=T_o}^{T_f} \delta^{T_f-s}\left(y_s - GaR_{i,s}(10\%)\right) \times \left(\tau - \mathbb{1}_{y_s<GaR_{i,s}(10\%)}\right), \qquad (12)$$

where $w_{i,t-h_d}$ is the weight corresponding to the individual $GaR_{i,s}(10\%)$ measure based on the high-frequency indicator i, which depends on the discounted TL given by $\lambda_{i,t-h_d}$, with discount factor $\delta = 0.9$ and $\kappa = 1$. Importantly, $s = T_o$ is the point at which the first prediction is computed, and $s = T_f$ is the point at which the most recent prediction can be evaluated with the high-frequency indicator up to the latest available observation. By using this framework, we can compute a combined $GaR(10\%)$ measure for each model.

3.4. GaR evaluation

We evaluate tail risk forecasts using a battery of indicators developed in the literature on forecast and risk management. Our main tool for assessing $GaR(10\%)$ point forecasts is the average TL, which has been shown to be particularly suitable when the object of interest is the forecast of a certain quantile of the dependent variable's conditional distribution (see Giacomini and Komunjer, 2005, Gneiting and Raftery 2007; Gneiting and Ranjan 2011, Manzan, 2015). In particular, Carriero et al. (2022) use this loss function to evaluate the predictive capacity of their models for quantifying tail risks.

The average TL for $\tau = 0.10$ is specified as follows,

$$TL_{\tau=10\%} = \frac{1}{T}\sum_{t=1}^{T}(y_t - GaR_t(10\%)) \times \left(\tau - \mathbb{1}_{y_t<GaR_t(10\%)}\right), \qquad (13)$$

where y_t is the actual GDP growth, $GaR_t(10\%)$ is the 10% predictive quantile of GDP growth, and the indicator function $\mathbb{1}_{(y_t<GaR_t(10\%))}$ takes a value of 1 if it is below the 10% forecast quantile and 0 otherwise.

Following the convention in the discipline (see Corradi and Swanson, 2006; Clark and McCracken, 2013) we use the Diebold and Mariano (1995) test (DM)[9], to assess the relative forecasting accuracy of our GaR models. In all the cases the models compared are non-nested. In the recent literature, Andreou et al. (2013), Pettenuzzo et al. (2016) and Carriero et al. (2022) followed the same approach.

In addition, we employ two coverage tests commonly used in the risk management literature to assess interval forecasts. Following Christoffersen (1998), the problem of assessing the adequacy of a VaR model can be reduced to the problem of determining whether the indicator of excess sequence (i.e., the $\mathbb{1}_{y_t<GaR_t(10\%)}$) has two properties: i) an unconditional coverage property, and ii) an independence property. In this setting, GaR forecasts are evaluated using the TL, which is a loss function generally used to assess the accuracy of VaR predictions (Giacomini and Komunjer, 2005). We evaluate these two conditions with the Unconditional Coverage (UC), and the Dynamic Quantile (DQ) tests (Engle and Manganelli, 2004), respectively. Specifically, the DQ is estimated using four lags of the excess sequence indicator (see Engle and Manganelli, 2004). Brownlees and Souza (2021) follow a similar approach for a multi-country GaR evaluation. Note that these two criteria can be achieved by more than one model, so at the end the TL is used in the final selection of the best performing model.

4. Empirical analysis

This section presents the statistical details for each model's computation and the results of the nowcasting exercise.

4.1. Parametrization and computational approach

[9] We use the variance adjustment of Harvey et al. (1997), which is supported by the results in Clark and McCracken (2013).

We consider different high-frequency GaR measures spanning from different variables and models. Below we discuss the choice of our parameters for each model:

- MIDAS-Q: Eq. (6) is estimated through the quantile approach described in Section 3.2.1 considering a third degree Almon lag polynomial ($c = 3$) with two end-point restrictions ($r = 2$), such that the number of parameters of the high-frequency indicator is reduced substantially to $c + 1 - r$. This lag structure has nice economic properties as it slowly decays towards zero (see Mogliani and Simoni, 2021).
- BMIDAS-Q: Based on the abovementioned constrained Almon lag structure for the high-frequency variable, Eq. (7) is estimated through the Bayesian methodology considered in Section 3.2.2. Specifically, the model considers standard uninformative priors on the coefficient vector to have a mean equal to 0 and a variance where all elements in the diagonal are equal to 9, except for the autoregressive lag of GDP, whose prior mean and variance are set to 0.5 and 0.1, respectively. Also, the scale and shape parameters of the inverse gamma function are set to 0.01. The Gibbs sampler is used to estimate the model parameters with 10,000 repetitions (for computation efficiency), after a burn-in period of 1,000 iterations, using the normal approximation, which simplifies the algorithm (Yang et al., 2015)[10]. The choice of these parameters closely resembles the ones of Ferrara et al. (2022), which is a natural benchmark model for our work.
- LASSO-Q: Following the model presented in Section 3.2.3, we set the penalty parameter λ equal to the 0.9 quantile of the pivotal distribution (see Belloni and Chernozhukov, 2011). Figure A1 shows the selected lags for LASSO-Q using each high-frequency predictor. Interestingly, historically LASSO-Q not only tends to select the most recent

[10] See Kozumi and Kobayashi (2011) for details on the estimation procedure.

daily lag of the given quarter, as one could expect, but others from past quarters. This is a key difference of this technique compared to MIDAS-Q and BMIDAS-Q, as the latter models have an arbitrary decaying weighting structure.

- EN-Q: Based on the model presented in Section 3.2.3, $\lambda_{1,\tau}$ is set as the penalty parameter of the LASSO-Q model, defined as above, and $\lambda_{2,\tau}$ is obtained by minimizing the mean cross-validated errors of the model, with the EN mixing parameter set to $\alpha = 0.5$ (Friedman et al., 2010). Figure A2 shows the selected lags for EN-Q using each high-frequency predictor. Analogous to LASSO-Q, we observe that this model historically selects different daily lags for each high-frequency variable.
- LASSO-PCA-Q and EN-PCA-Q: Based on the non-zero high-frequency lags selected with either LASSO-Q or EN-Q, we estimate factors by principal components analysis, select the optimal number of factors using the eigen ratio (Ahn and Horenstein, 2013), and keep the factors with associated p-values lower than 0.01 (or the statistically most significant ones). The final step of this procedure is estimated through the quantile approach described in Section 3.1.
- ASGL-Q: Based on the model presented in Section 3.2.5, we consider the following parametrization procedure. First, we carry out cross validation over different values of λ and α to get their optimal values. By estimating this model with the full sample and all the high-frequency variables (except the CISS, which is used as a benchmark), we obtain the optimal values of $\lambda = 0.010$ and $\alpha = 0.25$, which minimize the TL function. Second, we compute recursively both LASSO weights and group LASSO weights based on the regression on a subset of principal components. As suggested by Mendez-Civieta et al. (2021) this method achieves better results in terms of prediction error and stability of the variables selected when used in real datasets.

For each of these models[11], we construct the individual $GaR(10\%)$ nowcasts by estimating the 10% quantile forecast $\widehat{Q_{\tau=10\%}}\left(y_T|y_{T-1}, \boldsymbol{X}^{\boldsymbol{D}}_{\boldsymbol{i,T-h_d}}\right)$ conditional on one lag of GDP growth and the respective high-frequency indicator, as described in Table 1. This measure is computed recursively on a daily basis for each specification including a high-frequency indicator $\boldsymbol{X}^{\boldsymbol{D}}_{\boldsymbol{i,T-h_d}}$. The estimation sample spans from 1986Q1 to 2020Q4, and the daily nowcasts start on January 1, 2007.

Then, we use the forecast combination approach explained in Section 3.3 to summarize the information steaming from different daily indicators. By using the combination weights (see Eq. (12)), we compute the combined $GaR(10\%)$ recursively for each model (except for the ASGL-Q) as follows:

$$GaR^*_T(10\%) = \sum_i w_{i,T-h_d} \times \widehat{Q_{\tau=10\%}}\left(y_T|y_{T-1}, \boldsymbol{X}^{\boldsymbol{D}}_{\boldsymbol{i,T-h_d}}\right). \tag{14}$$

Eq. (14) allows us to capture the relative importance of individual $GaR(10\%)$ estimates and to deal with the potential problem of introducing many series, potentially correlated, into a common framework. It is important to stress that the combined $GaR(10\%)$ does not include the CISS, as it is the benchmark financial composite indicator. Figure A3 provides a recursive plot of the combination weights assigned to the various models using forecast combination approach. We observe that both, high-frequency real and financial indicators are important to produce accurate $GaR(10\%)$ nowcasts and the importance of each one is time varying.

[11] In the case of the AGSL-Q model, as it allows for multiple groups of variables, we compute the conditional 10% quantile using all the high-frequency indicators (except the CISS, which is the benchmark), so it produces directly a combined GaR forecast.

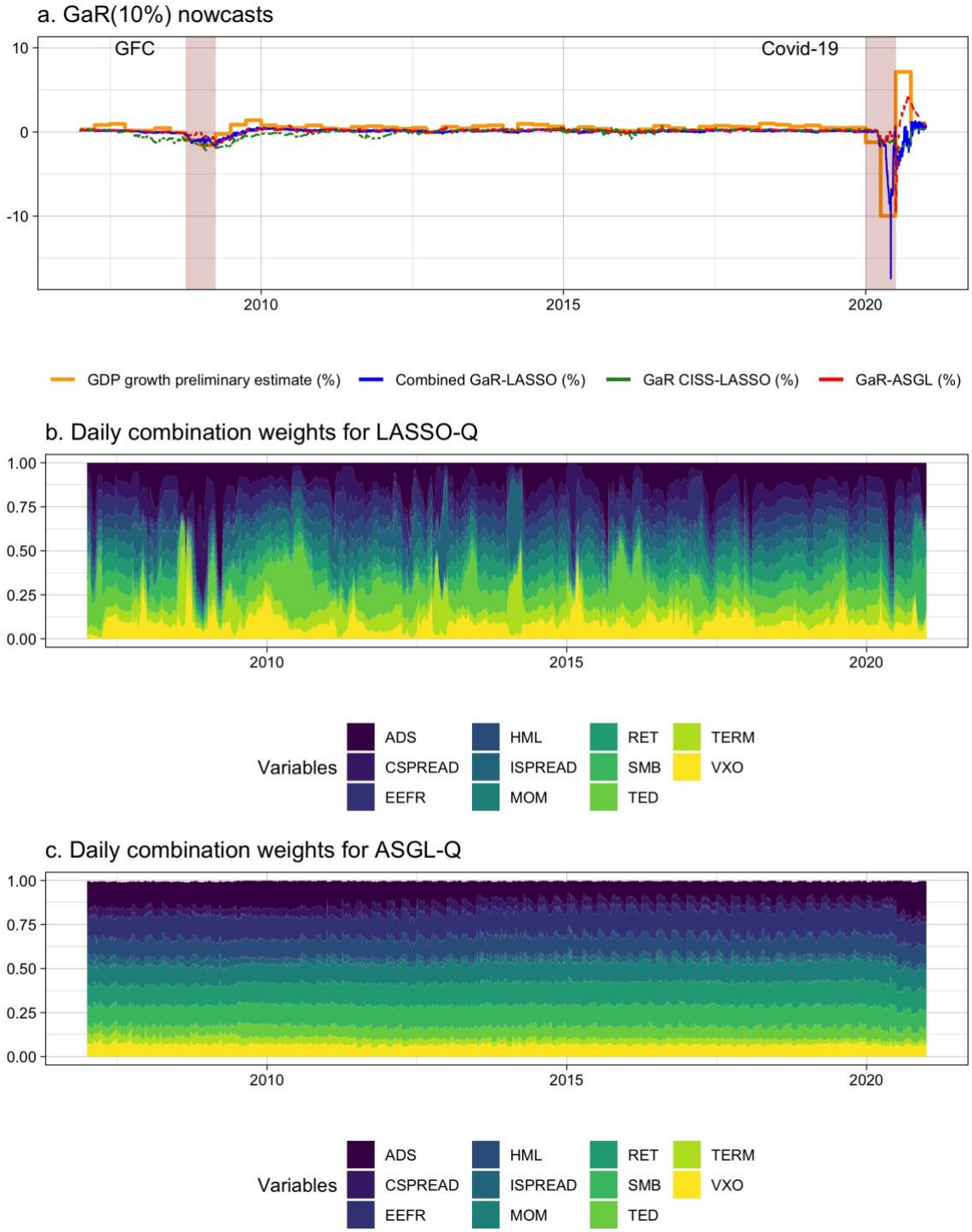
4.2. Nowcasting GaR

We recursively estimate all the above-mentioned specifications for each quarter from 1986Q1 to 2006Q4 and construct daily GaR nowcasts in pseudo-real-time starting from January 1, 2007. As a starting point, we show the combined $GaR(10\%)$ forecasts of our LASSO-Q model[12] compared to two alternative specifications, the individual $GaR(10\%)$ using the CISS and the combined $GaR(10\%)$ estimated by ASGL-Q. Recall that while LASSO-Q uses forecast combination as explained in Section 3.3, ASGL-Q uses all the information directly in the estimation. Figure 1a shows the preliminary estimates in real-time of the quarterly growth rate of the US along with the combined $GaR(10\%)$ and the two alternative models. Overall, we observe that the large negative growth rates that appear during recession periods, such as the GFC in 2008–2009 and the Covid-19 pandemic (that started in 2020), are captured effectively by our combined $GaR(10\%)$, while it is not the case for the last event with the alternative specifications. First, the difference in prediction between LASSO-Q and ASGL-Q are partially due to the difference in their combination schemes (see Figures 1b and 1c). While the weights of the former model are more volatile, those of the latter are more stable. Interestingly, in both frameworks, at the onset of the Covid-19 pandemic a larger weight is assigned to the ADS indicator, highlighting the usefulness of using real indicators in a GaR framework. Second, relative to the individual GaR using the CISS and the standard GaR framework that uses only a composite financial condition index (Adrian et al., 2019), a key aspect of our framework is that it allows for wider range of indicators which improves the accuracy of our predictions. Figure 1b provides a clearer view of this point by presenting the daily combination weights assigned to

[12] We chose to show the combined GaR nowcasts of the LASSO-Q model as, in general, performs relatively well compared to the rest of the models in terms of the average TL (see Table 2). Results for the other models are available upon request.

the different individual GaR nowcasts. We can observe that the relative importance of real and financial indicators is time varying. Indeed, we see that around the GFC that started in 2008, the ADS, VXO and CSPREAD indicators receive a relatively high weight across all models. In contrast, on the onset of the Covid-19 pandemic, all models assign higher weights to the ADS. On one hand, the former agrees with the general consensus in the macro-financial literature that highlights the financial nature of the GFC, in which financial intermediaries and markets amplified the shocks to the real economy (Isohätälä et al., 2016; Brunnermeier and Sannikov, 2016; Gertler and Gilchrist, 2018). On the other hand, the latter results recognize that the Covid-19 crisis was a product of the supply restrictions imposed to contain the pandemic, which were real and supply-side in nature (Guerrieri et al., 2022). Consequently, our estimated optimal weights suggest that it is fundamental to include real and financial daily indicators to improve GaR nowcasts. Interestingly, related works by Ferrara et al. (2022) and De Santis and Van der Veken (2020), which only include daily financial variables in the GaR framework, fail to capture the real magnitude of the risks during the Covid-19 period. This evidence indicates that financial variables alone play only a modest role in gauging the effect of this last recession.

Figure 1. GaR results for LASSO-Q and ASGL-Q



Source: Authors' computations.
Note: The estimation sample spans from 1986Q1 to 2020Q4, and the daily nowcasts start on January 1, 2007. In Panel a, the red shaded area represents NBER recessions at the end of the period. In Panel c, we omit the weight of the lagged GDP growth as it is close to zero.

4.3. Evaluation

In this section we perform the following exercises to assess the relative performance of: i) combined GaR nowcasts based on real and financial indicators compared to individual GaR nowcasts using a financial condition index (CISS) as a benchmark; and ii) individual GaR nowcasts of financial and real indicators relative to the combined GaR nowcasts, in the spirit of Figueres and Jarociński (2020).

4.3.1. Combined GaR (using financial and real variables) versus standard GaR

Table 2 reports the relative average TL of a given combined GaR model compared to the benchmark model (an individual GaR using the CISS), joint with its DM test probability value. A value lower than one in the TL implies that the combined model outperforms the benchmark, while for the DM test the alternative hypothesis is that the indicated forecast is more accurate than the benchmark (a rejection of the null is preferred). Notably, most models outperform the benchmark and rejects the hypothesis of equality of forecasts according to the DM test with 10% probability, which gives strong evidence of the benefits of combining multiple real and financial indicators into the GaR framework. Remarkably, LASSO-Q model tends to provide lower average TL and often rejects the null hypothesis of the DM test for different daily horizons, for both periods (including and not the Covid-19). We provide evidence that LASSO lag selection improves forecast accuracy while imposing fewer restrictions than traditional MIDAS models.

Table 3 reports the tests commonly used in the financial risk management literature, namely the UC and the DQ tests, to assess interval forecasts for the different combined GaR models. Specifically, the UC test reports the probability of the null hypothesis that the exceedance is equal to the quantile (non-rejection is better); and the DQ test the probability that the null

hypothesis is that the exceedance indicator is an i.i.d. process (non-rejection is better). Overall, models that use LASSO or EN do it better than MIDAS in this adequacy tests, with higher number of not rejections at 10% probability. These results hold for both, the period before Covid-19 and the period including it.

Table 2. Out-of-sample forecast accuracy based on the relative TL

Model:	$h_d = 0$		$h_d = 10$		$h_d = 20$		$h_d = 40$		$h_d = 60$	
	TL	**DM**	**TL**	**DM**	**TL**	**DM**	**TL**	**DM**	**TL**	**DM**
Panel A. Before COVID-19 (2007Q1 to 2019Q4)										
$GaR_T^{*,MIDAS}$	0.641	**0.001**	0.655	**0.001**	0.653	**0.000**	0.686	**0.001**	0.683	**0.001**
$GaR_T^{*,BMIDAS}$	0.606	**0.000**	0.616	**0.000**	0.631	**0.001**	0.643	**0.001**	0.654	**0.001**
$GaR_T^{*,LASSO}$	0.590	**0.001**	0.559	**0.000**	0.569	**0.000**	0.769	0.145	0.843	0.232
$GaR_T^{*,EN}$	0.956	0.415	0.978	0.461	0.932	0.366	0.853	0.273	0.858	0.277
$GaR_T^{*,LASSO-PCA}$	0.617	**0.001**	0.638	**0.002**	0.706	**0.010**	0.830	0.225	0.857	0.266
$GaR_T^{*,EN-PCA}$	0.617	**0.001**	0.691	**0.010**	0.741	**0.039**	0.809	0.176	0.844	0.251
$GaR_T^{*,ASGL}$	1.102	0.646	1.037	0.559	0.983	0.471	0.945	0.419	1.221	0.744
Panel B. Including COVID-19 (2007Q1 to 2020Q4)										
$GaR_T^{*,MIDAS}$	0.855	**0.027**	0.82	**0.005**	0.804	**0.022**	0.558	**0.094**	0.943	0.201
$GaR_T^{*,BMIDAS}$	0.878	**0.021**	0.849	**0.000**	0.839	**0.005**	0.558	**0.087**	0.932	0.141
$GaR_T^{*,LASSO}$	0.864	**0.002**	0.773	**0.006**	0.458	**0.096**	0.501	0.121	0.895	**0.092**
$GaR_T^{*,EN}$	0.953	0.243	0.969	0.330	0.822	0.153	0.563	0.139	0.917	0.173
$GaR_T^{*,LASSO-PCA}$	0.940	0.263	0.733	**0.041**	0.488	0.116	0.593	0.133	0.927	0.120
$GaR_T^{*,EN-PCA}$	0.911	0.102	0.850	**0.013**	0.841	**0.064**	0.691	0.116	0.903	0.123
$GaR_T^{*,ASGL}$	1.106	0.790	1.002	0.506	1.027	0.614	1.056	0.687	1.085	0.766

Note: This table shows the TL for each combined GaR relative to the individual GaR considering the CISS, for different daily horizons. We also report the p-values of the DM test for the null hypothesis of equality of forecasts, conducted on a one-sided basis, such that the alternative hypothesis is that the indicated forecast is more accurate than the benchmark (a rejection of the null is preferred). If the p-value is below 0.10 (bold values), we conclude that the forecast from a combined GaR model is more accurate than the benchmark.

Table 3. Out-of-sample forecast accuracy based on coverage tests

Model:	$h_d = 0$		$h_d = 10$		$h_d = 20$		$h_d = 40$		$h_d = 60$	
	UC	**DQ**	**UC**	**DQ**	**UC**	**DQ**	**UC**	**DQ**	**UC**	**DQ**
Panel A. Before COVID-19 (2007Q1 to 2019Q4)										
$GaR_T^{*,MIDAS}$	0.001	**0.619**	0.019	**0.164**	0.019	**0.144**	0.019	**0.141**	0.001	**0.619**
$GaR_T^{*,BMIDAS}$	0.001	**0.619**	0.001	**0.619**	0.001	**0.619**	0.001	**0.619**	0.001	**0.619**
$GaR_T^{*,LASSO}$	0.019	**0.849**	**0.273**	0.014	**0.273**	0.018	0.095	**0.590**	0.019	**0.272**
$GaR_T^{*,EN}$	**0.273**	**0.180**	**0.273**	**0.218**	**0.926**	**0.126**	**0.273**	0.045	**0.273**	**0.107**
$GaR_T^{*,LASSO-PCA}$	0.095	**0.316**	0.095	**0.344**	**0.273**	**0.378**	0.095	**0.630**	0.095	0.011
$GaR_T^{*,EN-PCA}$	0.019	**0.842**	**0.565**	0.021	**0.273**	**0.386**	0.095	**0.603**	0.095	**0.631**
$GaR_T^{*,ASGL}$	**0.427**	0.013	**0.226**	0.071	**0.226**	0.044	**0.926**	**0.200**	**0.226**	**0.493**
Panel B. Including COVID-19 (2007Q1 to 2020Q4)										
$GaR_T^{*,MIDAS}$	**0.208**	0.001	**0.455**	0.045	**0.455**	0.085	**0.455**	0.024	**0.208**	0.003
$GaR_T^{*,BMIDAS}$	0.068	0.040	0.068	0.063	0.068	0.080	0.068	0.080	0.068	0.080
$GaR_T^{*,LASSO}$	0.068	**0.917**	**0.786**	0.000	**0.786**	0.021	**0.786**	0.042	**0.208**	**0.266**
$GaR_T^{*,EN}$	**0.786**	0.007	**0.786**	0.009	**0.547**	0.029	**0.786**	0.042	**0.786**	0.076
$GaR_T^{*,LASSO-PCA}$	**0.455**	**0.235**	**0.455**	0.009	**0.786**	0.036	**0.455**	**0.118**	**0.455**	0.008
$GaR_T^{*,EN-PCA}$	**0.208**	**0.468**	**0.547**	0.000	**0.860**	**0.202**	**0.455**	**0.130**	**0.455**	**0.677**
$GaR_T^{*,ASGL}$	**0.160**	0.015	0.031	0.000	0.031	0.000	**0.312**	**0.237**	**0.031**	**0.371**

Note: This table shows the following two interval tests for different combined GaR: the unconditional coverage test (UC) of Kupiec (1995), which reports the probability of the null hypothesis that the exceedance is equal to the quantile (not rejecting is better); and the dynamic quantile test (DQ) of Engle and Manganelli (2004), where the null hypothesis is that the exceedance indicator is an i.i.d. process (non-rejection of the null is preferred). Bold values indicate that model passes the test with 10% probability.

4.3.2. Combined GaR (using financial and real variables) versus individual GaR

Now we address the question whether combined or individual indicators provide more accurate nowcasts, in the spirit of Figueres and Jarociński (2020). Table 4 reports the relative average TL of an individual GaR relative to the combined GaR (benchmark) using LASSO-Q, joint with its Diebold-Mariano (DM) probability test. Again, a value lower than one in the TL implies that the individual GaR outperforms the benchmark (combined GaR), while for the DM the alternative hypothesis is that the indicated forecast is more accurate than the benchmark. In this setting, we would prefer to have a TL greater than 1 and not reject the null hypothesis, to have evidence of our combined GaR framework. Overall, results suggest that with 10% probability we cannot reject the null hypothesis of the DM test, meaning that our combined GaR is more accurate. Nonetheless, the individual GaR specification using the ADS indicator is the only one that has a relative TL lower than one for daily horizons greater or equal to 10 days and for the period including the Covid-19 pandemic. These results suggest that individual GaR models that include the ADS perform relatively better than those that do not include it, as in Pettenuzzo et al. (2016) and Lima et al. (2020). Also, this result recognizes that the Covid-19 crisis was a product of the supply restrictions imposed to contain the pandemic, which were real and supply-side in nature (Guerrieri et al., 2022). Results for alternative models are presented in Appendix B.

Table 4. LASSO-Q: Out-of-sample forecast accuracy based on the relative TL

Model:	$h_d = 0$		$h_d = 10$		$h_d = 20$		$h_d = 40$		$h_d = 60$	
	TL	**DM**	**TL**	**DM**	**TL**	**DM**	**TL**	**DM**	**TL**	**DM**
Panel A. Before COVID-19 (2007Q1 to 2019Q4)										
$GaR_T^{ISPREAD}$	2.232	**0.978**	2.072	**0.990**	1.768	**0.988**	1.772	**0.997**	1.692	**0.996**
GaR_T^{EEFR}	2.000	**0.965**	2.392	**0.935**	1.819	**0.972**	1.711	**0.971**	1.649	**0.968**
GaR_T^{RET}	2.107	**0.973**	2.049	**0.962**	1.792	**0.946**	2.200	**0.994**	1.594	**0.971**
GaR_T^{SMB}	2.079	**0.991**	2.202	**0.991**	1.833	**0.994**	1.930	**0.986**	2.069	**0.990**
GaR_T^{HML}	2.200	**0.957**	1.447	**0.916**	1.570	**0.973**	1.514	**0.968**	1.655	**0.916**
GaR_T^{MOM}	1.448	**0.987**	2.076	**0.959**	1.578	**0.987**	2.072	**0.960**	1.756	**0.973**
GaR_T^{VXO}	1.813	**0.991**	1.550	**0.995**	1.521	**0.997**	1.258	**0.921**	1.268	**0.886**
$GaR_T^{CSPREAD}$	2.242	**0.979**	1.789	**0.993**	1.367	**1.000**	1.275	**0.963**	1.259	**0.951**
GaR_T^{TERM}	2.198	**0.984**	1.946	**0.991**	1.728	**0.989**	1.754	**0.990**	1.684	**0.989**
GaR_T^{TED}	1.643	**0.918**	1.603	**0.968**	1.485	**0.964**	1.440	**0.960**	1.397	**0.931**
GaR_T^{ADS}	1.678	**0.950**	1.496	**0.910**	1.175	**0.717**	0.939	**0.375**	1.263	**0.915**
Panel B. Including COVID-19 (2007Q1 to 2020Q4)										
$GaR_T^{ISPREAD}$	1.445	**0.968**	1.555	**0.995**	1.530	**0.990**	1.837	**0.973**	1.346	**0.983**
GaR_T^{EEFR}	1.449	**0.982**	1.707	**0.992**	1.550	**0.987**	1.855	**0.963**	1.367	**0.979**
GaR_T^{RET}	1.408	**0.992**	1.556	**0.989**	1.503	**0.976**	1.776	**0.964**	1.208	**0.957**
GaR_T^{SMB}	1.271	**0.948**	1.510	**0.989**	1.301	**0.991**	1.667	**0.960**	1.302	**0.996**
GaR_T^{HML}	1.453	**0.995**	1.504	**0.983**	1.281	**0.913**	1.834	**0.942**	1.300	**0.975**
GaR_T^{MOM}	1.274	**0.981**	1.712	**0.971**	1.510	**0.969**	1.722	**0.934**	1.307	**0.988**
GaR_T^{VXO}	1.196	**0.908**	1.335	**0.995**	1.317	**0.994**	1.426	**0.893**	1.129	**0.950**
$GaR_T^{CSPREAD}$	1.336	**0.991**	1.351	**0.993**	1.280	**0.939**	1.501	**0.888**	1.133	**0.956**
GaR_T^{TERM}	1.42	**0.974**	1.502	**0.995**	1.432	**0.993**	1.789	**0.971**	1.334	**0.989**
GaR_T^{TED}	1.315	**0.960**	1.433	**0.994**	1.420	**0.985**	1.731	**0.950**	1.279	**0.980**
GaR_T^{ADS}	1.375	**0.923**	0.595	**0.174**	0.655	**0.159**	0.504	**0.152**	0.743	**0.260**

Note: This table shows the TL for each individual GaR forecast relative to the combined GaR forecast, for different daily horizons. We also report the p-values of the DM test for the null hypothesis of equality of forecasts, conducted on a one-sided basis, such that the alternative hypothesis is that the indicated forecast is more accurate than the combined GaR (non-rejection of the null is preferred). If the p-value is above 0.10 (bold values), we conclude that the forecast from the combined GaR is more accurate than that of the individual GaR.

Table 5 reports the UC and the DQ tests, to assess interval forecasts for the different individual GaR specifications estimated by LASSO-Q. Again, the UC test reports the probability of the null hypothesis that the exceedance is equal to the quantile (not rejecting is better); and the DQ test the probability that the null hypothesis is that the exceedance indicator is an i.i.d. process (not rejecting is better). The evidence of individual GaR specifications is mixed and depends on the daily horizon. Results for alternative models are presented in Appendix C.

Overall, the evidence we present here supports the time-varying importance of both daily financial and real indicators for estimating GaR. Our results are consistent with those for the Eurozone reported by Ferrara et al. (2022) and for the US reported by De Santis and Van der Veken (2020), in that daily financial variables provide policymakers with timely warnings about the downside risks of GDP. Nevertheless, we provide further and clearer evidence in the way suggested by Pettenuzzo et al. (2016), emphasizing the usefulness of incorporating a real high-frequency indicator in the forecasting regressions, such as the ADS. Furthermore, by comparing our combined GaR against the standard GaR using a financial condition index, namely the CISS, we show that our framework performs significantly better. This is also true when comparing our combined GaR framework against different individual GaR, although the ones that only include the ADS perform similarly when considered the Covid-19 period. This evidence ultimately suggests that financial variables alone played a limited role in gauging the downside risk for the GDP of the Covid-19 pandemic and acknowledges the complex ways in which real and financial variables interconnect and determine economic growth, which is causal in nature.

Table 5. LASSO-Q: Out-of-sample forecast accuracy based on coverage tests

Model:	$h_d = 0$		$h_d = 10$		$h_d = 20$		$h_d = 40$		$h_d = 60$	
	UC	**DQ**	**UC**	**DQ**	**UC**	**DQ**	**UC**	**DQ**	**UC**	**DQ**
Panel A. Before COVID-19 (2007Q1 to 2019Q4)										
$GaR_T^{ISPREAD}$	**0.926**	**0.161**	**0.926**	0.072	**0.926**	**0.121**	**0.565**	0.070	**0.565**	**0.052**
GaR_T^{EEFR}	**0.717**	**0.131**	**0.226**	0.012	**0.427**	0.070	**0.226**	0.089	**0.107**	0.014
GaR_T^{RET}	**0.226**	0.001	0.001	0.000	0.046	0.000	0.046	0.000	0.001	0.000
GaR_T^{SMB}	0.046	0.001	**0.427**	0.015	**0.226**	**0.103**	0.006	0.001	**0.107**	0.002
GaR_T^{HML}	0.046	0.050	**0.107**	0.000	**0.226**	0.001	0.018	0.006	0.018	0.003
GaR_T^{MOM}	**0.427**	**0.781**	**0.717**	**0.146**	0.046	0.099	0.046	0.000	**0.226**	0.003
GaR_T^{VXO}	**0.107**	0.066	0.046	**0.812**	0.018	0.063	0.046	0.067	0.046	**0.707**
$GaR_T^{CSPREAD}$	**0.226**	0.036	**0.226**	0.038	**0.717**	**0.741**	**0.717**	**0.799**	**0.565**	**0.921**
GaR_T^{TERM}	**0.565**	0.087	**0.565**	0.088	**0.565**	0.080	**0.926**	**0.280**	**0.273**	**0.701**
GaR_T^{TED}	**0.226**	0.011	**0.107**	0.003	**0.107**	0.003	0.046	0.003	0.046	0.023
GaR_T^{ADS}	**0.107**	0.067	0.018	0.050	**0.107**	0.043	**0.226**	**0.186**	**0.226**	**0.230**
Panel B. Including COVID-19 (2007Q1 to 2020Q4)										
$GaR_T^{ISPREAD}$	**0.312**	0.003	**0.312**	0.002	**0.312**	0.002	**0.547**	0.001	**0.547**	0.001
GaR_T^{EEFR}	**0.160**	0.007	0.031	0.000	0.074	0.001	0.031	0.002	0.012	0.000
GaR_T^{RET}	0.074	0.000	0.000	0.000	0.004	0.000	0.004	0.000	0.000	0.000
GaR_T^{SMB}	0.012	0.001	**0.160**	0.004	0.074	0.083	0.001	0.000	0.031	0.001
GaR_T^{HML}	0.004	0.000	0.012	0.000	0.031	0.001	0.001	0.000	0.004	0.002
GaR_T^{MOM}	**0.160**	**0.425**	**0.160**	0.018	0.004	0.001	0.004	0.000	0.074	0.000
GaR_T^{VXO}	0.031	**0.338**	0.004	**0.267**	0.001	0.011	0.004	0.042	0.004	**0.215**
$GaR_T^{CSPREAD}$	0.031	0.005	0.031	0.011	**0.160**	**0.204**	**0.160**	**0.427**	**0.547**	**0.921**
GaR_T^{TERM}	**0.547**	0.002	**0.547**	0.001	**0.547**	0.002	**0.312**	0.008	**0.860**	0.021
GaR_T^{TED}	0.031	0.003	0.012	0.000	0.012	0.000	0.004	0.000	0.004	0.001
GaR_T^{ADS}	0.074	**0.137**	0.004	0.003	0.031	0.001	0.074	**0.230**	0.031	0.018

Note: This table shows the following two interval tests for different individual $GaR(10\%)$ models: the unconditional coverage test (UC) of Kupiec (1995), which reports the probability of the null hypothesis that the exceedance is equal to the quantile (not rejecting is better); and the dynamic quantile test (DQ) of Engle and Manganelli (2004), where the null hypothesis is that the exceedance indicator is an i.i.d. process (non-rejection of the null is preferred). Bold values indicate that model passes the test with 10% probability.

5. Conclusions

We show that on a daily frequency both real and financial variables provide valuable information for monitoring vulnerable periods of economic activity. Our key contribution is that we show that by incorporating these two types of variables simultaneously in the GaR framework, we can provide an early signal of GDP downturns in pseudo real-time that works well for both, the GFC and the Covid-19 episodes.

Our flexible approach allows us to emphasize the important role of economic theory and economic intuition when the results of forecast combination exercises are interpreted and to improve the point forecast itself. By acknowledging the complexity of the nowcasting task in macroeconomics, especially when using high frequency data, we contribute to a better understanding of the economic signals that can be extracted from this daily information to anticipate economic declines. In this sense, we show that during the GFC and the Covid-19 pandemic, the optimal forecasting weights of real and financial variables clearly changed. In the former period, financial indicators were fundamental, such as credit spreads and the VXO; however, they failed to capture the magnitude of the GDP decline observed on the onset of the Covid-19 pandemic. This, in turn, is explained by the different natures of the two crises, that we can only grasp because we understand (up to some extent) the economic mechanisms behind these two crises.

Interestingly, within the set of financial variables, VXO and CSPREAD are especially relevant across models around the GFC, which highlights the prominent role of uncertainty in determining economic outcomes. However, as mentioned above, the financial indicators alone were unable to forecast GDP low quantiles during Covid-19. Indeed, only by including the ADS

index we managed to gauge both the sign and the magnitude of the downside GDP risk in this period.

We showed that our combined GaR model outperforms the standard GaR model, which considers only financial condition indicators (Adrian et al., 2019; Ferrara et al. 2022). More technically, we have assessed this by: i) comparing the performance of combined GaR nowcasts to individual GaR nowcasts using the CISS as a benchmark; and ii) individual GaR nowcasts using different financial and real indicators relative to the combined GaR nowcasts. Our implementation uses different dimension reduction techniques such as MIDAS and shrinkage.

We have compared seven different models and 12 high-frequency predictors using a forecast combination approach with time-varying optimal weights. In addition, we have used a novel approach based on adaptive sparse group LASSO for quantile regressions, which allows for multiple groups and sparsity within the high-frequency lagged vector (Mendez-Civieta et al., 2021). While this model has some nice properties for high-dimensional problems, we found that LASSO-Q tend to outperform the rest of models in terms of forecast accuracy at different daily horizons. This is probably a consequence of traditional MIDAS restrictions on the lag structure of the high frequency indicator, which do not necessarily improve forecast accuracy. Therefore, our results give further support to past evidence, in that shrinkage models should ideally be used to select the number of lags of the high-frequency predictors. Regarding the ASGL-Q model, it displays more stable weights than LASSO-Q, which could arguably be a potential cause of the difference in accuracy. Nonetheless, these two weighting schemes emphasize the importance of the ADS indicator for forecasting during the Covid-19 period.

We used a single indicator for capturing the role of real economic activity, namely the ADS, basically because it is the only indicator that is available at the daily frequency. We also introduced

weekly vintages of this indicator to perform the nowcasting exercise in pseudo-real time. Nevertheless, we believe that more indicators gauging the informational content of different facets of economic activity and the credit markets will prove to be fundamental in the future, not only to achieve greater forecasting accuracy in real time, but also to understand the causes of ongoing crises, even before the true causal mechanisms are clear to the professionals. In this sense, we see our models as a contribution in the direction of anticipating and understanding economic dangers while they are occurring.

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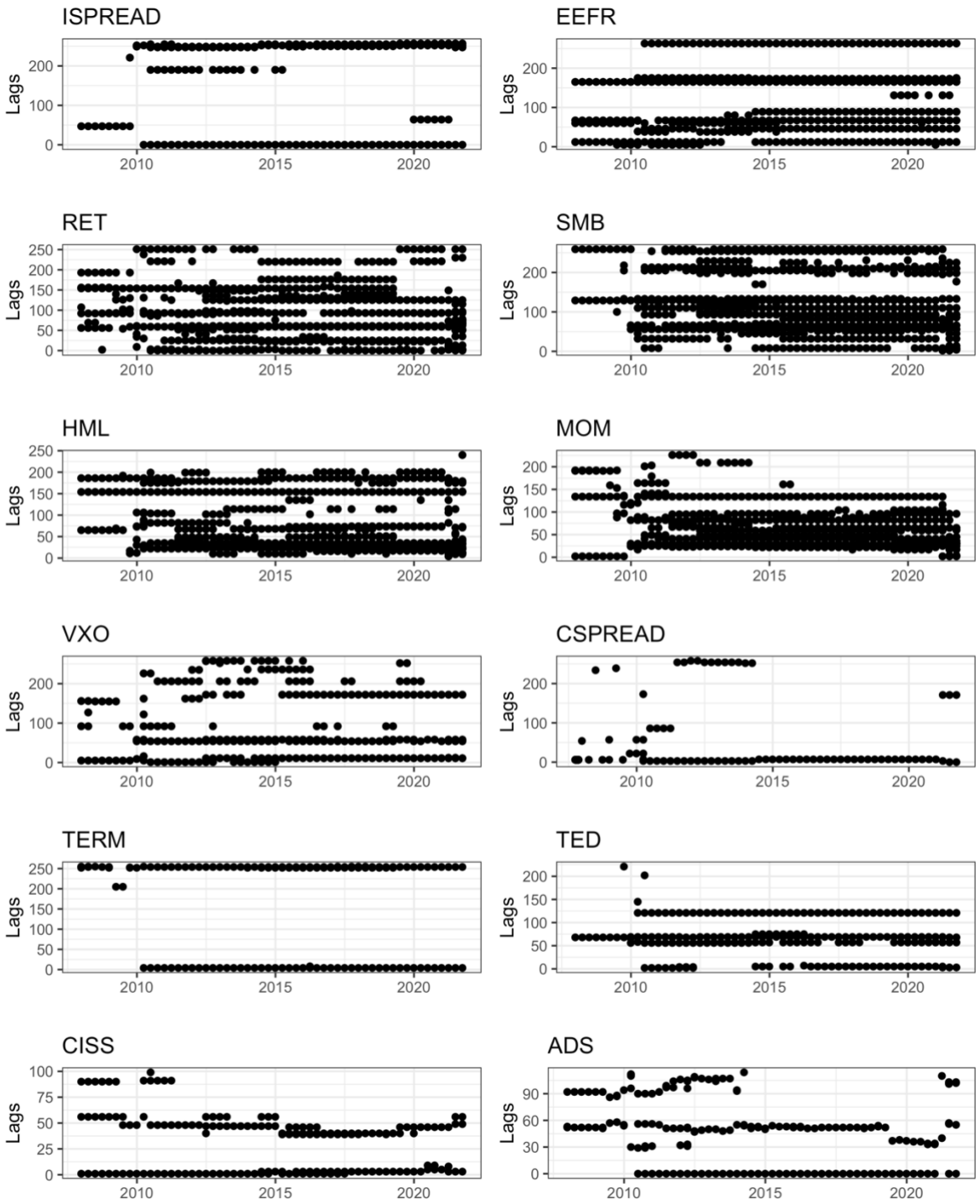
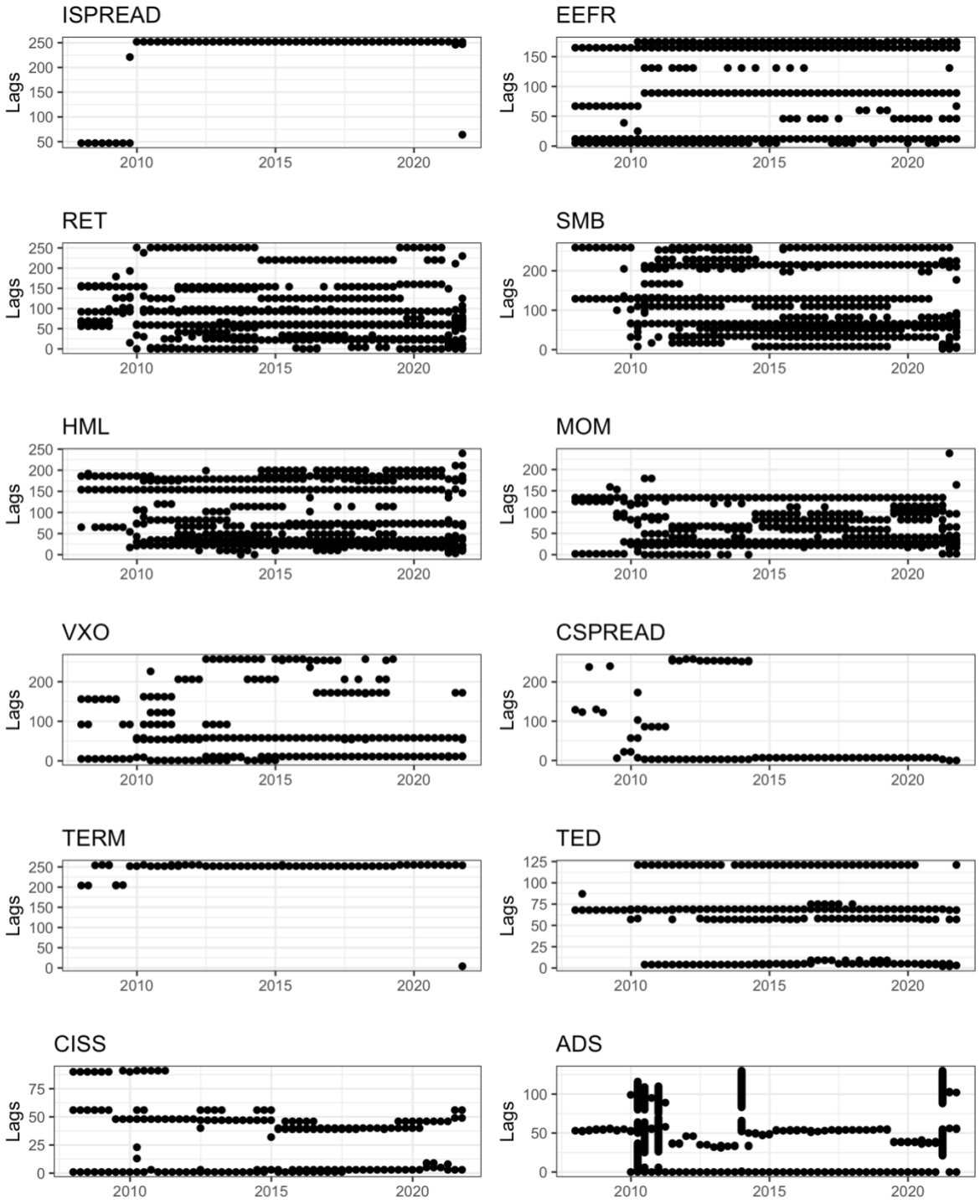
Appendix A

Figure A1. LASSO-Q selected high-frequency lags from individual GaR specifications



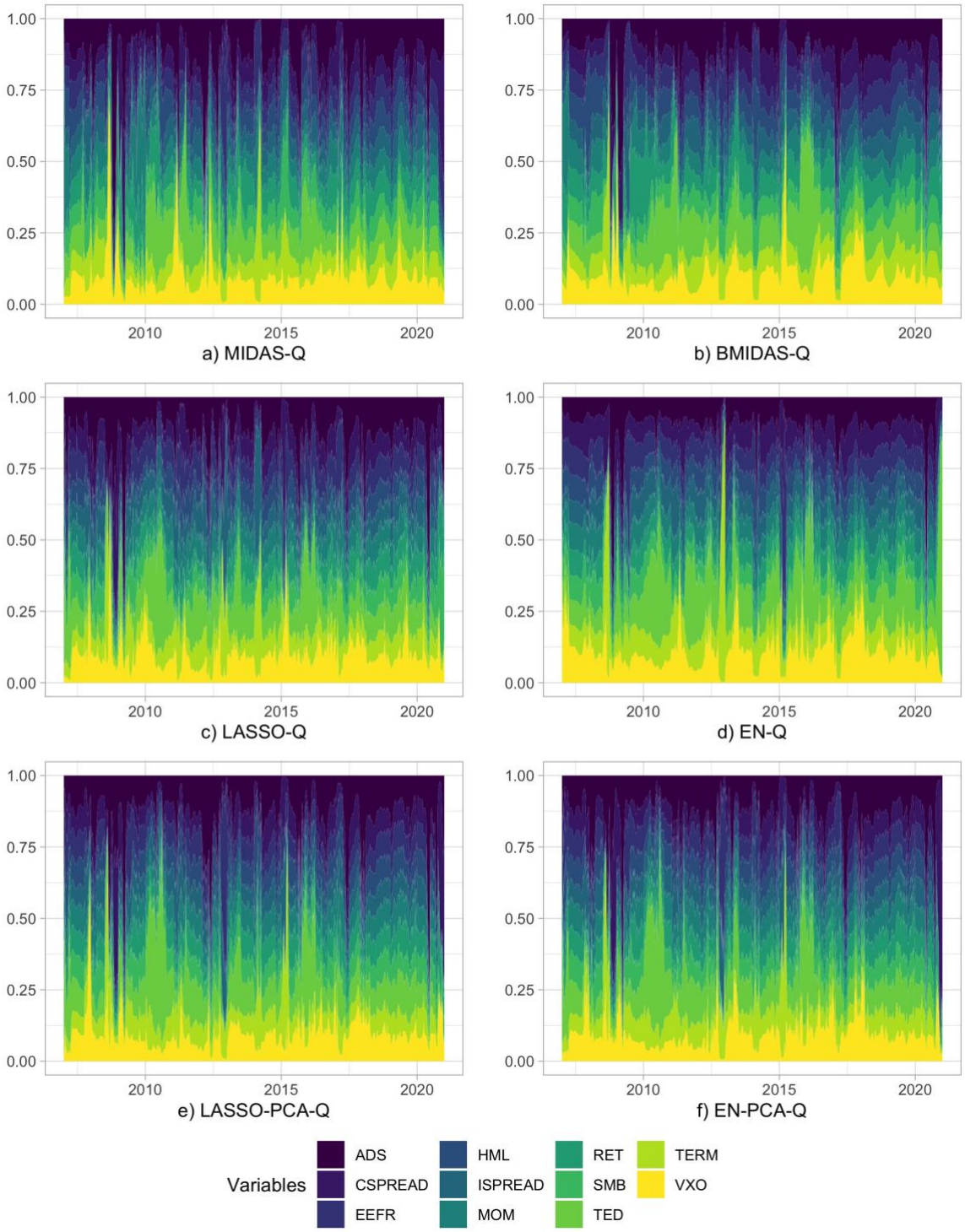
Source: Authors' computations.
Note: Figure shows the selected daily lags from LASSO-Q individual models that corresponds to the last day of each quarter.

Figure A2. EN-Q selected high-frequency lags from individual GaR specifications



Source: Authors' computations.
Note: Figure shows the selected daily lags from EN-Q individual models that corresponds to the last day of each quarter.

Figure A3. Daily forecast combination weights



Source: Authors' computations.
Note: The estimation sample spans from 1986Q1 to 2020Q4, and the daily nowcasts start in January 1, 2007.

Appendix B: TL for alternative models

Table B1. MIDAS-Q: Out-of-sample forecast accuracy based on the relative TL

Model:	$h_d = 0$		$h_d = 10$		$h_d = 20$		$h_d = 40$		$h_d = 60$	
	TL	DM	TL	DM	TL	DM	TL	DM	TL	DM
Panel A. Before COVID-19 (2007Q1 to 2019Q4)										
$GaR_T^{ISPREAD}$	2.338	**0.990**	2.285	**0.990**	2.313	**0.99**	2.212	**0.991**	2.234	**0.987**
GaR_T^{EEFR}	2.317	**0.971**	2.185	**0.962**	2.221	**0.967**	2.12	**0.968**	1.999	**0.955**
GaR_T^{RET}	1.446	**0.989**	1.506	**0.994**	1.595	**0.997**	1.808	**0.996**	1.637	**0.996**
GaR_T^{SMB}	2.11	**0.988**	2.124	**0.991**	2.07	**0.993**	2.008	**0.992**	2.058	**0.989**
GaR_T^{HML}	2.141	**0.987**	2.204	**0.986**	2.29	**0.978**	2.452	**0.980**	2.169	**0.981**
GaR_T^{MOM}	2.097	**0.978**	1.987	**0.978**	1.945	**0.979**	1.752	**0.978**	1.780	**0.974**
GaR_T^{VXO}	1.617	**0.951**	1.589	**0.959**	1.537	**0.975**	1.493	**0.986**	1.467	**0.994**
$GaR_T^{CSPREAD}$	2.052	**0.996**	1.834	**0.997**	1.579	**1.000**	1.363	**0.997**	1.414	**1.000**
GaR_T^{TERM}	2.313	**0.989**	2.253	**0.989**	2.252	**0.988**	2.173	**0.991**	2.198	**0.987**
GaR_T^{TED}	1.957	**0.982**	1.929	**0.978**	1.889	**0.977**	1.758	**0.980**	1.784	**0.981**
GaR_T^{ADS}	1.474	**0.936**	1.527	**0.953**	1.377	**0.928**	1.100	**0.698**	1.433	**0.908**
Panel B. Including COVID-19 (2007Q1 to 2020Q4)										
$GaR_T^{ISPREAD}$	1.437	**0.996**	1.470	**0.998**	1.489	**0.992**	2.201	**0.952**	1.334	**0.991**
GaR_T^{EEFR}	1.327	**0.988**	1.375	**0.984**	1.418	**0.969**	2.087	**0.923**	1.231	**0.969**
GaR_T^{RET}	1.370	**0.977**	1.460	**0.984**	1.494	**0.981**	2.235	**0.951**	1.327	**0.954**
GaR_T^{SMB}	1.491	**0.990**	1.539	**0.993**	1.537	**0.985**	2.267	**0.951**	1.355	**0.985**
GaR_T^{HML}	1.585	**0.989**	1.650	**0.996**	1.633	**0.997**	2.433	**0.966**	1.433	**0.996**
GaR_T^{MOM}	1.664	**0.970**	1.698	**0.975**	1.729	**0.972**	2.453	**0.959**	1.511	**0.948**
GaR_T^{VXO}	1.202	**0.935**	1.251	**0.961**	1.300	**0.960**	1.918	**0.927**	1.107	**0.892**
$GaR_T^{CSPREAD}$	1.300	**0.995**	1.307	**0.991**	1.270	**0.946**	1.819	**0.871**	1.115	**0.984**
GaR_T^{TERM}	1.438	**0.995**	1.471	**0.997**	1.486	**0.990**	2.18	**0.948**	1.318	**0.990**
GaR_T^{TED}	1.407	**0.994**	1.454	**0.993**	1.465	**0.984**	2.113	**0.935**	1.276	**0.987**
GaR_T^{ADS}	1.089	**0.796**	0.788	**0.247**	0.762	**0.248**	0.764	**0.231**	0.978	**0.446**

Note: This table shows the TL for each combined GaR relative to the individual GaR considering the CISS, for different daily horizons. We also report the p-values of the DM test for the null hypothesis of equality of forecasts, conducted on a one-sided basis, such that the alternative hypothesis is that the indicated forecast is more accurate than the benchmark (a rejection of the null is preferred). If the p-value is below 0.10 (bold values), we conclude that the forecast from a combined GaR model is more accurate than the benchmark.

Table B2. BMIDAS-Q: Out-of-sample forecast accuracy based on the relative TL

Model:	$h_d = 0$		$h_d = 10$		$h_d = 20$		$h_d = 40$		$h_d = 60$	
	TL	**DM**	**TL**	**DM**	**TL**	**DM**	**TL**	**DM**	**TL**	**DM**
Panel A. Before COVID-19 (2007Q1 to 2019Q4)										
$GaR_T^{ISPREAD}$	2.133	**0.972**	2.109	**0.971**	2.075	**0.970**	2.054	**0.970**	2.026	**0.967**
GaR_T^{EEFR}	1.892	**0.969**	1.842	**0.965**	1.802	**0.964**	1.799	**0.965**	1.685	**0.956**
GaR_T^{RET}	1.259	**0.966**	1.348	**0.989**	1.331	**0.975**	1.333	**0.969**	1.346	**0.973**
GaR_T^{SMB}	2.217	**0.978**	2.215	**0.979**	2.160	**0.981**	2.181	**0.977**	2.152	**0.977**
GaR_T^{HML}	2.15	**0.965**	2.227	**0.965**	2.265	**0.954**	2.432	**0.948**	2.058	**0.942**
GaR_T^{MOM}	2.706	**0.979**	2.546	**0.983**	2.325	**0.984**	2.101	**0.972**	2.100	**0.980**
GaR_T^{VXO}	1.378	**0.959**	1.360	**0.965**	1.335	**0.970**	1.311	**0.971**	1.245	**0.957**
$GaR_T^{CSPREAD}$	1.561	**1.000**	1.544	**1.000**	1.528	**1.000**	1.519	**1.000**	1.493	**1.000**
GaR_T^{TERM}	2.163	**0.973**	2.128	**0.971**	2.089	**0.970**	2.068	**0.970**	2.056	**0.969**
GaR_T^{TED}	1.488	**0.917**	1.458	**0.916**	1.421	**0.914**	1.395	**0.930**	1.374	**0.935**
GaR_T^{ADS}	1.177	**0.908**	1.197	**0.935**	1.125	**0.884**	0.961	**0.330**	1.119	**0.784**
Panel B. Including COVID-19 (2007Q1 to 2020Q4)										
$GaR_T^{ISPREAD}$	1.342	**0.989**	1.368	**0.991**	1.367	**0.977**	2.054	**0.923**	1.245	**0.972**
GaR_T^{EEFR}	1.232	**0.982**	1.248	**0.983**	1.26	**0.960**	1.909	**0.910**	1.135	**0.934**
GaR_T^{RET}	1.183	**0.951**	1.26	**0.969**	1.248	**0.947**	1.834	**0.877**	1.117	**0.980**
GaR_T^{SMB}	1.392	**0.993**	1.414	**0.991**	1.403	**0.975**	2.150	**0.928**	1.295	**0.983**
GaR_T^{HML}	1.435	**0.983**	1.491	**0.990**	1.496	**0.985**	2.268	**0.945**	1.304	**0.975**
GaR_T^{MOM}	1.51	**0.991**	1.488	**0.995**	1.445	**0.994**	2.094	**0.933**	1.285	**0.990**
GaR_T^{VXO}	1.039	**0.720**	1.067	**0.894**	1.112	**0.927**	1.704	**0.892**	1.009	**0.545**
$GaR_T^{CSPREAD}$	1.194	**0.991**	1.227	**0.998**	1.229	**0.985**	1.850	**0.902**	1.127	**1.000**
GaR_T^{TERM}	1.354	**0.990**	1.377	**0.991**	1.376	**0.977**	2.073	**0.925**	1.26	**0.977**
GaR_T^{TED}	1.242	**0.969**	1.263	**0.969**	1.257	**0.935**	1.885	**0.882**	1.142	**0.964**
GaR_T^{ADS}	1.000	**0.501**	0.754	**0.204**	0.742	**0.215**	0.689	**0.196**	0.907	**0.305**

Note: This table shows the TL for each combined GaR relative to the individual GaR considering the CISS, for different daily horizons. We also report the p-values of the DM test for the null hypothesis of equality of forecasts, conducted on a one-sided basis, such that the alternative hypothesis is that the indicated forecast is more accurate than the benchmark (a rejection of the null is preferred). If the p-value is below 0.10 (bold values), we conclude that the forecast from a combined GaR model is more accurate than the benchmark.

Table B3. EN-Q: Out-of-sample forecast accuracy based on the relative TL

Model:	$h_d = 0$		$h_d = 10$		$h_d = 20$		$h_d = 40$		$h_d = 60$	
	TL	**DM**	**TL**	**DM**	**TL**	**DM**	**TL**	**DM**	**TL**	**DM**
Panel A. Before COVID-19 (2007Q1 to 2019Q4)										
$GaR_T^{ISPREAD}$	1.339	**0.988**	1.327	**0.992**	1.305	**0.991**	1.462	**0.981**	1.456	**0.976**
GaR_T^{EEFR}	1.373	**0.959**	1.569	**0.919**	1.41	**0.956**	1.564	**0.969**	1.549	**0.959**
GaR_T^{RET}	1.589	**0.992**	1.536	**0.971**	1.919	**0.992**	1.732	**0.98**	2.263	**0.986**
GaR_T^{SMB}	1.308	**1.000**	1.362	**1.000**	1.611	**0.992**	1.756	**0.997**	1.681	**0.996**
GaR_T^{HML}	1.631	**0.991**	1.182	**0.925**	1.289	**0.993**	1.854	**0.997**	1.809	**0.993**
GaR_T^{MOM}	1.050	**0.808**	1.355	**0.950**	1.312	**0.989**	1.870	**0.984**	1.450	**0.951**
GaR_T^{VXO}	1.315	**0.974**	1.328	**0.979**	1.293	**0.970**	1.257	**0.890**	1.284	**0.924**
$GaR_T^{CSPREAD}$	1.170	**0.907**	1.177	**0.934**	1.144	**0.911**	1.303	**0.961**	1.250	**0.955**
GaR_T^{TERM}	1.273	**0.987**	1.255	**0.993**	1.253	**0.994**	1.427	**0.974**	1.398	**0.971**
GaR_T^{TED}	1.200	**0.830**	1.195	**0.853**	1.181	**0.855**	1.364	**0.915**	1.261	**0.845**
GaR_T^{ADS}	1.038	**0.589**	1.038	**0.588**	0.96	**0.405**	0.865	**0.230**	1.048	**0.672**
Panel B. Including COVID-19 (2007Q1 to 2020Q4)										
$GaR_T^{ISPREAD}$	1.165	**0.967**	1.151	**0.988**	1.33	**0.955**	1.926	**0.906**	1.202	**0.993**
GaR_T^{EEFR}	1.216	**0.981**	1.261	**0.977**	1.382	**0.953**	1.983	**0.910**	1.159	**0.976**
GaR_T^{RET}	1.414	**0.988**	1.381	**0.988**	1.649	**0.972**	1.992	**0.911**	1.414	**0.991**
GaR_T^{SMB}	1.203	**0.980**	1.429	**0.989**	1.409	**0.986**	2.108	**0.921**	1.268	**0.992**
GaR_T^{HML}	1.367	**0.994**	1.436	**0.980**	1.179	**0.975**	2.634	**0.951**	1.266	**0.996**
GaR_T^{MOM}	1.222	**0.952**	1.374	**0.982**	1.439	**0.949**	2.247	**0.931**	1.196	**0.973**
GaR_T^{VXO}	1.078	**0.926**	1.127	**0.992**	1.264	**0.931**	1.839	**0.877**	1.135	**0.959**
$GaR_T^{CSPREAD}$	1.152	**0.952**	1.132	**0.972**	1.293	**0.877**	1.845	**0.872**	1.114	**0.928**
GaR_T^{TERM}	1.145	**0.945**	1.157	**0.985**	1.35	**0.958**	1.988	**0.918**	1.187	**0.989**
GaR_T^{TED}	1.140	**0.911**	1.155	**0.960**	1.316	**0.903**	1.927	**0.885**	1.132	**0.923**
GaR_T^{ADS}	0.961	**0.380**	0.809	**0.169**	0.672	**0.146**	0.733	**0.190**	0.845	**0.316**

Note: This table shows the TL for each combined GaR relative to the individual GaR considering the CISS, for different daily horizons. We also report the p-values of the DM test for the null hypothesis of equality of forecasts, conducted on a one-sided basis, such that the alternative hypothesis is that the indicated forecast is more accurate than the benchmark (a rejection of the null is preferred). If the p-value is below 0.10 (bold values), we conclude that the forecast from a combined GaR model is more accurate than the benchmark.

Table B4. LASSO-PCA-Q: Out-of-sample forecast accuracy based on the relative TL

Model:	$h_d = 0$		$h_d = 10$		$h_d = 20$		$h_d = 40$		$h_d = 60$	
	TL	**DM**	**TL**	**DM**	**TL**	**DM**	**TL**	**DM**	**TL**	**DM**
Panel A. Before COVID-19 (2007Q1 to 2019Q4)										
$GaR_T^{ISPREAD}$	2.200	**0.978**	2.260	**0.988**	1.868	**0.988**	1.723	**0.997**	1.678	**0.996**
GaR_T^{EEFR}	2.254	**0.961**	2.295	**0.957**	2.074	**0.973**	1.795	**0.974**	1.822	**0.976**
GaR_T^{RET}	2.222	**0.980**	2.270	**0.968**	1.777	**0.927**	2.197	**0.984**	1.544	**0.978**
GaR_T^{SMB}	2.288	**0.992**	1.851	**0.971**	1.826	**0.988**	1.649	**0.981**	1.588	**0.989**
GaR_T^{HML}	2.163	**0.940**	2.425	**0.957**	1.938	**0.960**	2.188	**0.988**	1.726	**0.942**
GaR_T^{MOM}	1.475	**0.995**	1.958	**0.95**	1.622	**0.988**	1.976	**0.971**	1.700	**0.989**
GaR_T^{VXO}	1.785	**0.990**	1.714	**0.994**	1.591	**0.999**	1.328	**0.940**	1.220	**0.867**
$GaR_T^{CSPREAD}$	2.271	**0.982**	1.939	**0.996**	1.406	**1.000**	1.241	**0.915**	1.227	**0.931**
GaR_T^{TERM}	2.204	**0.986**	2.105	**0.987**	1.816	**0.987**	1.730	**0.992**	1.656	**0.990**
GaR_T^{TED}	1.644	**0.923**	1.736	**0.965**	1.562	**0.965**	1.407	**0.951**	1.379	**0.933**
GaR_T^{ADS}	1.187	**0.847**	1.186	**0.834**	0.928	**0.307**	0.943	**0.374**	1.228	**0.884**
Panel B. Including COVID-19 (2007Q1 to 2020Q4)										
$GaR_T^{ISPREAD}$	1.418	**0.964**	1.807	**0.994**	2.591	**0.952**	2.139	**0.957**	1.319	**0.977**
GaR_T^{EEFR}	1.502	**0.969**	1.938	**0.988**	2.834	**0.958**	2.284	**0.955**	1.445	**0.975**
GaR_T^{RET}	1.392	**0.992**	1.743	**0.974**	2.423	**0.909**	2.07	**0.947**	1.171	**0.981**
GaR_T^{SMB}	1.239	**0.908**	1.602	**0.952**	2.201	**0.926**	1.857	**0.893**	1.227	**0.995**
GaR_T^{HML}	1.314	**0.971**	2.051	**0.99**	2.658	**0.964**	3.143	**0.963**	1.209	**0.968**
GaR_T^{MOM}	1.14	**0.882**	1.990	**0.983**	2.444	**0.912**	2.059	**0.926**	1.208	**0.992**
GaR_T^{VXO}	1.087	**0.700**	1.480	**0.983**	2.122	**0.921**	1.688	**0.881**	1.090	**0.963**
$GaR_T^{CSPREAD}$	1.271	**0.963**	1.532	**0.983**	2.139	**0.888**	1.725	**0.874**	1.079	**0.981**
GaR_T^{TERM}	1.377	**0.956**	1.742	**0.995**	2.385	**0.940**	2.091	**0.952**	1.300	**0.984**
GaR_T^{TED}	1.274	**0.932**	1.662	**0.988**	2.450	**0.934**	2.018	**0.93**	1.247	**0.978**
GaR_T^{ADS}	1.499	**0.851**	0.606	**0.139**	0.610	**0.113**	0.605	**0.122**	0.902	**0.301**

Note: This table shows the TL for each combined GaR relative to the individual GaR considering the CISS, for different daily horizons. We also report the p-values of the DM test for the null hypothesis of equality of forecasts, conducted on a one-sided basis, such that the alternative hypothesis is that the indicated forecast is more accurate than the benchmark (a rejection of the null is preferred). If the p-value is below 0.10 (bold values), we conclude that the forecast from a combined GaR model is more accurate than the benchmark.

Table B5. EN-PCA-Q: Out-of-sample forecast accuracy based on the relative TL

Model:	$h_d = 0$		$h_d = 10$		$h_d = 20$		$h_d = 40$		$h_d = 60$	
	TL	**DM**	**TL**	**DM**	**TL**	**DM**	**TL**	**DM**	**TL**	**DM**
Panel A. Before COVID-19 (2007Q1 to 2019Q4)										
$GaR_T^{ISPREAD}$	2.232	**0.978**	2.072	**0.990**	1.768	**0.988**	1.772	**0.997**	1.692	**0.996**
GaR_T^{EEFR}	2.000	**0.965**	2.392	**0.935**	1.819	**0.972**	1.711	**0.971**	1.649	**0.968**
GaR_T^{RET}	2.107	**0.973**	2.049	**0.962**	1.792	**0.946**	2.200	**0.994**	1.594	**0.971**
GaR_T^{SMB}	2.079	**0.991**	2.202	**0.991**	1.833	**0.994**	1.930	**0.986**	2.069	**0.990**
GaR_T^{HML}	2.200	**0.957**	1.447	**0.916**	1.57	**0.973**	1.514	**0.968**	1.655	**0.916**
GaR_T^{MOM}	1.448	**0.987**	2.076	**0.959**	1.578	**0.987**	2.072	**0.960**	1.756	**0.973**
GaR_T^{VXO}	1.813	**0.991**	1.550	**0.995**	1.521	**0.997**	1.258	**0.921**	1.268	**0.886**
$GaR_T^{CSPREAD}$	2.242	**0.979**	1.789	**0.993**	1.367	**1.000**	1.275	**0.963**	1.259	**0.951**
GaR_T^{TERM}	2.198	**0.984**	1.946	**0.991**	1.728	**0.989**	1.754	**0.990**	1.684	**0.989**
GaR_T^{TED}	1.643	**0.918**	1.603	**0.968**	1.485	**0.964**	1.44	**0.960**	1.397	**0.931**
GaR_T^{ADS}	1.678	**0.950**	1.496	**0.910**	1.175	**0.717**	0.939	**0.375**	1.263	**0.915**
Panel B. Including COVID-19 (2007Q1 to 2020Q4)										
$GaR_T^{ISPREAD}$	1.445	**0.968**	1.555	**0.995**	1.530	**0.99**	1.837	**0.973**	1.346	**0.983**
GaR_T^{EEFR}	1.449	**0.982**	1.707	**0.992**	1.550	**0.987**	1.855	**0.963**	1.367	**0.979**
GaR_T^{RET}	1.408	**0.992**	1.556	**0.989**	1.503	**0.976**	1.776	**0.964**	1.208	**0.957**
GaR_T^{SMB}	1.271	**0.948**	1.510	**0.989**	1.301	**0.991**	1.667	**0.960**	1.302	**0.996**
GaR_T^{HML}	1.453	**0.995**	1.504	**0.983**	1.281	**0.913**	1.834	**0.942**	1.300	**0.975**
GaR_T^{MOM}	1.274	**0.981**	1.712	**0.971**	1.510	**0.969**	1.722	**0.934**	1.307	**0.988**
GaR_T^{VXO}	1.196	**0.908**	1.335	**0.995**	1.317	**0.994**	1.426	**0.893**	1.129	**0.950**
$GaR_T^{CSPREAD}$	1.336	**0.991**	1.351	**0.993**	1.280	**0.939**	1.501	**0.888**	1.133	**0.956**
GaR_T^{TERM}	1.420	**0.974**	1.502	**0.995**	1.432	**0.993**	1.789	**0.971**	1.334	**0.989**
GaR_T^{TED}	1.315	**0.960**	1.433	**0.994**	1.420	**0.985**	1.731	**0.950**	1.279	**0.980**
GaR_T^{ADS}	1.375	**0.923**	0.595	**0.174**	0.655	**0.159**	0.504	**0.152**	0.743	**0.260**

Note: This table shows the TL for each combined GaR relative to the individual GaR considering the CISS, for different daily horizons. We also report the p-values of the DM test for the null hypothesis of equality of forecasts, conducted on a one-sided basis, such that the alternative hypothesis is that the indicated forecast is more accurate than the benchmark (a rejection of the null is preferred). If the p-value is below 0.10 (bold values), we conclude that the forecast from a combined GaR model is more accurate than the benchmark.

Appendix C: UC and DQ tests for alternative models

Table C1. MIDAS-Q: Out-of-sample forecast accuracy based on coverage tests

Model:	$h_d = 0$		$h_d = 10$		$h_d = 20$		$h_d = 40$		$h_d = 60$	
	UC	**DQ**	**UC**	**DQ**	**UC**	**DQ**	**UC**	**DQ**	**UC**	**DQ**
Panel A. Before COVID-19 (2007Q1 to 2019Q4)										
$GaR_T^{ISPREAD}$	**0.565**	0.039	**0.565**	0.042	**0.565**	0.032	**0.565**	0.034	**0.565**	0.032
GaR_T^{EEFR}	**0.273**	0.003	0.095	**0.142**	**0.273**	0.002	**0.273**	0.003	**0.273**	0.004
GaR_T^{RET}	**0.427**	**0.452**	**0.226**	**0.713**	0.046	**0.281**	0.046	0.027	0.046	0.015
GaR_T^{SMB}	**0.926**	**0.200**	**0.717**	**0.509**	**0.717**	**0.553**	**0.926**	**0.191**	**0.926**	0.153
GaR_T^{HML}	**0.565**	**0.560**	**0.565**	**0.561**	**0.717**	**0.506**	**0.717**	0.077	**0.926**	**0.273**
GaR_T^{MOM}	**0.926**	**0.142**	**0.926**	0.054	**0.565**	**0.147**	**0.273**	**0.368**	0.095	**0.141**
GaR_T^{VXO}	**0.926**	**0.903**	**0.717**	**0.878**	**0.717**	**0.867**	**0.717**	**0.863**	**0.226**	**0.887**
$GaR_T^{CSPREAD}$	**0.717**	0.088	**0.926**	0.090	**0.926**	**0.102**	**0.273**	**0.925**	0.095	**0.826**
GaR_T^{TERM}	**0.565**	0.056	**0.565**	0.060	**0.273**	0.002	**0.273**	0.002	**0.273**	0.002
GaR_T^{TED}	**0.226**	**0.273**	**0.107**	**0.265**	**0.226**	**0.309**	**0.226**	**0.310**	**0.226**	**0.323**
GaR_T^{ADS}	**0.226**	**0.229**	**0.226**	**0.201**	**0.226**	**0.230**	0.006	0.000	**0.107**	0.072
Panel B. Including COVID-19 (2007Q1 to 2020Q4)										
$GaR_T^{ISPREAD}$	**0.547**	0.001	**0.547**	0.001	**0.547**	0.000	**0.547**	0.000	**0.547**	0.000
GaR_T^{EEFR}	**0.860**	0.001	**0.786**	0.004	**0.860**	0.000	**0.860**	0.000	**0.860**	0.000
GaR_T^{RET}	0.074	0.059	0.031	0.015	0.004	0.005	0.004	0.001	0.004	0.001
GaR_T^{SMB}	**0.312**	0.004	**0.160**	0.014	**0.160**	0.015	**0.312**	0.003	**0.312**	0.003
GaR_T^{HML}	**0.547**	0.008	**0.547**	0.007	**0.160**	0.007	**0.160**	0.001	**0.312**	0.005
GaR_T^{MOM}	**0.312**	0.005	**0.312**	0.000	**0.547**	0.004	**0.860**	0.008	**0.786**	0.000
GaR_T^{VXO}	**0.547**	**0.675**	**0.160**	**0.763**	**0.160**	**0.685**	**0.160**	**0.656**	0.031	**0.584**
$GaR_T^{CSPREAD}$	**0.160**	0.100	**0.312**	0.035	**0.312**	0.056	**0.860**	**0.626**	**0.786**	**0.405**
GaR_T^{TERM}	**0.547**	0.001	**0.547**	0.001	**0.860**	0.000	**0.860**	0.000	**0.860**	0.000
GaR_T^{TED}	0.031	**0.242**	0.012	0.028	0.031	0.033	0.031	0.041	0.031	0.032
GaR_T^{ADS}	**0.160**	**0.146**	**0.160**	0.088	0.074	0.015	0.004	0.002	0.031	0.008

Note: This table shows the following two interval tests for different combined GaR: the unconditional coverage test (UC) of Kupiec (1995), which reports the probability of the null hypothesis that the exceedance is equal to the quantile (not rejecting is better); and the dynamic quantile test (DQ) of Engle and Manganelli (2004), where the null hypothesis is that the exceedance indicator is an i.i.d. process (non-rejection of the null is preferred). Bold values indicate that model passes the test with 10% probability.

Table C2. BMIDAS-Q: Out-of-sample forecast accuracy based on coverage tests

Model:	$h_d = 0$		$h_d = 10$		$h_d = 20$		$h_d = 40$		$h_d = 60$	
	UC	DQ	UC	DQ	UC	DQ	UC	DQ	UC	DQ
Panel A. Before COVID-19 (2007Q1 to 2019Q4)										
$GaR_T^{ISPREAD}$	0.095	**0.139**	0.095	**0.141**	0.095	**0.142**	0.095	**0.141**	**0.273**	0.012
GaR_T^{EEFR}	**0.273**	0.006	**0.273**	0.006	0.095	0.082	0.095	**0.109**	0.095	0.074
GaR_T^{RET}	**0.427**	0.058	**0.926**	**0.682**	**0.565**	**0.877**	**0.427**	**0.638**	**0.717**	**0.383**
GaR_T^{SMB}	**0.565**	0.000	**0.565**	0.000	**0.565**	0.000	**0.565**	0.000	**0.565**	0.000
GaR_T^{HML}	**0.565**	0.029	**0.565**	0.022	**0.273**	**0.228**	**0.273**	0.005	**0.273**	**0.139**
GaR_T^{MOM}	**0.273**	0.012	**0.273**	0.013	**0.273**	0.013	**0.273**	0.012	0.095	**0.125**
GaR_T^{VXO}	**0.565**	**0.907**	**0.565**	**0.908**	**0.565**	**0.909**	**0.926**	**0.907**	**0.717**	**0.710**
$GaR_T^{CSPREAD}$	0.095	**0.885**	0.095	**0.881**	0.095	**0.88**	0.095	**0.874**	0.019	0.811
GaR_T^{TERM}	0.095	**0.139**	0.095	**0.142**	0.095	**0.142**	0.095	**0.14**	**0.273**	0.012
GaR_T^{TED}	**0.717**	**0.731**	**0.717**	**0.713**	**0.717**	**0.695**	**0.717**	**0.686**	**0.717**	**0.696**
GaR_T^{ADS}	**0.565**	**0.892**	**0.926**	**0.901**	**0.565**	**0.758**	**0.095**	**0.432**	0.095	**0.375**
Panel B. Including COVID-19 (2007Q1 to 2020Q4)										
$GaR_T^{ISPREAD}$	**0.455**	0.015	**0.455**	0.015	**0.455**	0.015	**0.455**	0.014	**0.786**	0.004
GaR_T^{EEFR}	**0.786**	0.000	**0.786**	0.000	**0.455**	0.001	**0.455**	0.004	**0.455**	0.002
GaR_T^{RET}	**0.160**	**0.140**	**0.547**	**0.163**	**0.860**	**0.362**	**0.16**	**0.168**	**0.312**	0.052
GaR_T^{SMB}	**0.860**	0.000	**0.860**	0.000	**0.860**	0.000	**0.86**	0.000	**0.860**	0.000
GaR_T^{HML}	**0.860**	0.001	**0.860**	0.000	**0.786**	0.007	**0.786**	0.001	**0.786**	0.050
GaR_T^{MOM}	**0.786**	0.003	**0.786**	0.003	**0.786**	0.004	**0.786**	0.004	**0.455**	0.009
GaR_T^{VXO}	**0.786**	**0.864**	**0.860**	**0.462**	**0.860**	**0.68**	**0.547**	**0.709**	**0.312**	**0.119**
$GaR_T^{CSPREAD}$	**0.455**	**0.875**	**0.455**	**0.860**	**0.455**	**0.856**	**0.455**	**0.856**	**0.208**	**0.616**
GaR_T^{TERM}	**0.455**	0.015	**0.455**	0.015	**0.455**	0.015	**0.455**	0.015	**0.786**	0.004
GaR_T^{TED}	**0.312**	**0.328**	**0.312**	**0.327**	**0.312**	**0.32**	**0.312**	**0.315**	**0.312**	**0.324**
GaR_T^{ADS}	**0.860**	**0.716**	**0.547**	**0.529**	**0.547**	**0.835**	**0.208**	**0.569**	**0.455**	**0.372**

Note: This table shows the following two interval tests for different combined GaR: the unconditional coverage test (UC) of Kupiec (1995), which reports the probability of the null hypothesis that the exceedance is equal to the quantile (not rejecting is better); and the dynamic quantile test (DQ) of Engle and Manganelli (2004), where the null hypothesis is that the exceedance indicator is an i.i.d. process (non-rejection of the null is preferred). Bold values indicate that model passes the test with 10% probability.

Table C3. EN-Q: Out-of-sample forecast accuracy based on coverage tests

Model:	$h_d = 0$		$h_d = 10$		$h_d = 20$		$h_d = 40$		$h_d = 60$	
	UC	DQ	UC	DQ	UC	DQ	UC	DQ	UC	DQ
Panel A. Before COVID-19 (2007Q1 to 2019Q4)										
$GaR_T^{ISPREAD}$	**0.717**	**0.488**	**0.717**	**0.464**	**0.717**	**0.488**	**0.926**	**0.216**	**0.717**	**0.481**
GaR_T^{EEFR}	**0.926**	0.099	**0.565**	**0.147**	**0.926**	0.002	**0.926**	**0.124**	**0.926**	0.073
GaR_T^{RET}	**0.717**	0.007	**0.107**	0.000	**0.427**	0.000	**0.717**	0.002	**0.226**	0.010
GaR_T^{SMB}	**0.565**	0.071	**0.565**	0.064	**0.926**	**0.131**	**0.717**	0.012	**0.926**	0.021
GaR_T^{HML}	**0.107**	0.000	**0.717**	0.032	**0.926**	0.032	**0.427**	0.003	**0.427**	0.017
GaR_T^{MOM}	**0.565**	0.001	**0.273**	**0.648**	**0.107**	0.000	**0.427**	0.007	**0.427**	**0.134**
GaR_T^{VXO}	**0.427**	**0.656**	**0.427**	**0.645**	**0.226**	**0.358**	**0.427**	**0.583**	**0.926**	**0.922**
$GaR_T^{CSPREAD}$	**0.717**	**0.856**	**0.717**	**0.879**	**0.717**	**0.855**	**0.717**	**0.856**	**0.565**	**0.900**
GaR_T^{TERM}	**0.926**	**0.244**	**0.565**	**0.779**	**0.565**	**0.783**	**0.565**	**0.783**	**0.565**	**0.780**
GaR_T^{TED}	**0.107**	**0.330**	0.046	**0.235**	**0.107**	**0.329**	0.018	0.068	**0.107**	**0.521**
GaR_T^{ADS}	**0.565**	**0.741**	**0.926**	**0.678**	**0.565**	**0.573**	0.019	**0.286**	**0.565**	**0.889**
Panel B. Including COVID-19 (2007Q1 to 2020Q4)										
$GaR_T^{ISPREAD}$	**0.312**	**0.145**	**0.312**	**0.144**	**0.160**	0.028	**0.312**	0.007	**0.160**	0.015
GaR_T^{EEFR}	**0.547**	0.085	**0.86**	**0.313**	**0.547**	0.001	**0.547**	0.079	**0.547**	0.094
GaR_T^{RET}	**0.312**	0.000	0.031	0.000	**0.160**	0.000	**0.312**	0.003	0.074	0.001
GaR_T^{SMB}	**0.86**	0.011	**0.547**	0.000	**0.547**	0.038	**0.312**	0.002	**0.547**	0.016
GaR_T^{HML}	0.031	0.000	**0.160**	0.000	**0.547**	0.069	0.074	0.000	**0.160**	0.011
GaR_T^{MOM}	**0.86**	0.000	**0.860**	0.002	0.012	0.000	**0.160**	0.001	**0.160**	0.080
GaR_T^{VXO}	**0.312**	**0.498**	**0.160**	**0.111**	0.074	0.056	**0.160**	**0.113**	**0.547**	**0.458**
$GaR_T^{CSPREAD}$	**0.312**	**0.402**	**0.312**	**0.389**	**0.312**	**0.420**	**0.312**	**0.384**	**0.860**	**0.426**
GaR_T^{TERM}	**0.312**	0.014	**0.547**	**0.129**	**0.547**	0.088	**0.547**	0.074	**0.547**	0.072
GaR_T^{TED}	0.031	0.039	0.012	0.020	0.031	0.040	0.004	0.003	0.031	0.082
GaR_T^{ADS}	**0.786**	**0.921**	**0.547**	**0.551**	**0.547**	0.022	**0.208**	**0.432**	**0.547**	**0.776**

Note: This table shows the following two interval tests for different combined GaR: the unconditional coverage test (UC) of Kupiec (1995), which reports the probability of the null hypothesis that the exceedance is equal to the quantile (not rejecting is better); and the dynamic quantile test (DQ) of Engle and Manganelli (2004), where the null hypothesis is that the exceedance indicator is an i.i.d. process (non-rejection of the null is preferred). Bold values indicate that model passes the test with 10% probability.

Table C4. LASSO-PCA-Q: Out-of-sample forecast accuracy based on coverage tests

Model:	$h_d = 0$		$h_d = 10$		$h_d = 20$		$h_d = 40$		$h_d = 60$	
	UC	DQ	UC	DQ	UC	DQ	UC	DQ	UC	DQ
Panel A. Before COVID-19 (2007Q1 to 2019Q4)										
$GaR_T^{ISPREAD}$	**0.926**	**0.226**	**0.717**	**0.227**	**0.926**	**0.16**	**0.926**	0.069	**0.565**	0.050
GaR_T^{EEFR}	**0.565**	**0.394**	**0.926**	**0.110**	**0.926**	0.025	**0.926**	0.022	**0.717**	0.029
GaR_T^{RET}	**0.107**	0.050	**0.717**	**0.364**	**0.926**	**0.905**	0.006	0.000	0.018	0.007
GaR_T^{SMB}	**0.427**	**0.154**	**0.717**	**0.239**	**0.926**	**0.279**	**0.717**	**0.435**	**0.565**	**0.089**
GaR_T^{HML}	**0.107**	**0.123**	**0.427**	0.009	**0.427**	**0.133**	**0.226**	0.008	**0.427**	**0.336**
GaR_T^{MOM}	**0.226**	0.084	**0.926**	**0.884**	**0.427**	**0.626**	**0.926**	0.096	0.046	0.006
GaR_T^{VXO}	**0.107**	**0.481**	**0.226**	**0.570**	**0.226**	**0.382**	**0.107**	**0.189**	**0.717**	**0.711**
$GaR_T^{CSPREAD}$	**0.717**	**0.146**	**0.717**	**0.148**	**0.926**	**0.923**	**0.926**	**0.925**	**0.565**	**0.929**
GaR_T^{TERM}	**0.273**	0.003	**0.273**	0.003	**0.926**	**0.200**	**0.565**	0.079	**0.273**	**0.678**
GaR_T^{TED}	**0.427**	**0.469**	**0.226**	**0.180**	**0.226**	**0.180**	**0.107**	**0.185**	**0.107**	**0.450**
GaR_T^{ADS}	**0.226**	0.043	**0.427**	0.100	**0.226**	**0.085**	**0.226**	0.014	**0.226**	**0.229**
Panel B. Including COVID-19 (2007Q1 to 2020Q4)										
$GaR_T^{ISPREAD}$	**0.312**	0.004	**0.160**	0.004	**0.312**	0.001	**0.312**	0.001	**0.547**	0.001
GaR_T^{EEFR}	**0.547**	0.009	**0.312**	0.002	**0.312**	0.001	**0.312**	0.000	**0.160**	0.002
GaR_T^{RET}	0.031	0.001	**0.312**	0.046	**0.312**	0.035	0.000	0.000	0.004	0.007
GaR_T^{SMB}	**0.160**	0.009	**0.160**	0.007	**0.312**	0.028	**0.160**	0.028	**0.547**	0.000
GaR_T^{HML}	0.031	0.082	0.074	0.000	0.074	0.005	0.031	0.003	0.074	0.041
GaR_T^{MOM}	0.031	0.038	**0.312**	0.031	0.074	**0.136**	**0.547**	0.037	0.004	0.000
GaR_T^{VXO}	0.031	0.066	0.074	0.163	0.074	0.074	0.031	0.023	**0.312**	**0.372**
$GaR_T^{CSPREAD}$	**0.312**	**0.436**	**0.312**	**0.566**	**0.547**	**0.925**	**0.547**	**0.928**	**0.860**	**0.954**
GaR_T^{TERM}	**0.860**	0.000	**0.860**	0.000	**0.312**	0.006	**0.547**	0.001	**0.860**	0.014
GaR_T^{TED}	0.074	0.018	0.031	0.002	0.031	0.001	0.012	0.001	0.012	0.015
GaR_T^{ADS}	0.074	0.001	**0.312**	0.095	0.074	0.023	**0.160**	0.021	0.031	0.004

Note: This table shows the following two interval tests for different combined GaR: the unconditional coverage test (UC) of Kupiec (1995), which reports the probability of the null hypothesis that the exceedance is equal to the quantile (not rejecting is better); and the dynamic quantile test (DQ) of Engle and Manganelli (2004), where the null hypothesis is that the exceedance indicator is an i.i.d. process (non-rejection of the null is preferred). Bold values indicate that model passes the test with 10% probability.

Table C5. EN-PCA-Q: Out-of-sample forecast accuracy based on coverage tests

Model:	$h_d = 0$		$h_d = 10$		$h_d = 20$		$h_d = 40$		$h_d = 60$	
	UC	DQ	UC	DQ	UC	DQ	UC	DQ	UC	DQ
Panel A. Before COVID-19 (2007Q1 to 2019Q4)										
$GaR_T^{ISPREAD}$	**0.717**	**0.422**	**0.717**	**0.256**	**0.926**	**0.159**	**0.926**	0.061	**0.565**	0.054
GaR_T^{EEFR}	**0.565**	**0.769**	**0.427**	0.060	**0.926**	**0.114**	**0.427**	**0.455**	**0.427**	**0.589**
GaR_T^{RET}	**0.427**	0.042	**0.427**	**0.507**	**0.226**	0.061	0.006	0.000	0.006	0.002
GaR_T^{SMB}	**0.427**	**0.607**	0.046	0.001	**0.226**	0.035	**0.226**	**0.219**	**0.926**	0.009
GaR_T^{HML}	**0.107**	**0.115**	**0.717**	**0.333**	**0.926**	**0.253**	**0.427**	**0.120**	**0.717**	**0.155**
GaR_T^{MOM}	**0.226**	0.083	**0.717**	0.092	**0.226**	**0.716**	**0.427**	0.003	**0.226**	0.021
GaR_T^{VXO}	**0.226**	**0.614**	**0.107**	**0.188**	**0.226**	**0.383**	**0.107**	**0.337**	**0.427**	**0.725**
$GaR_T^{CSPREAD}$	**0.717**	**0.147**	**0.717**	**0.148**	**0.926**	**0.919**	**0.926**	**0.925**	**0.565**	**0.930**
GaR_T^{TERM}	**0.565**	0.062	**0.273**	0.003	**0.926**	**0.198**	**0.565**	0.074	**0.273**	**0.680**
GaR_T^{TED}	**0.427**	**0.469**	**0.226**	**0.180**	**0.226**	**0.180**	**0.107**	**0.185**	**0.107**	**0.451**
GaR_T^{ADS}	**0.107**	0.031	**0.226**	0.096	**0.226**	0.006	**0.226**	0.008	**0.226**	**0.390**
Panel B. Including COVID-19 (2007Q1 to 2020Q4)										
$GaR_T^{ISPREAD}$	**0.160**	0.010	**0.160**	0.005	**0.312**	0.001	**0.312**	0.001	**0.547**	0.001
GaR_T^{EEFR}	**0.547**	0.026	0.074	0.002	**0.312**	0.003	0.074	0.011	0.074	0.018
GaR_T^{RET}	**0.160**	0.005	0.074	0.002	0.031	0.000	0.000	0.000	0.001	0.000
GaR_T^{SMB}	**0.160**	**0.260**	0.004	0.000	0.031	0.000	0.031	0.009	**0.312**	0.000
GaR_T^{HML}	0.012	0.001	**0.160**	0.003	**0.312**	0.049	0.074	0.014	**0.160**	0.012
GaR_T^{MOM}	0.031	**0.254**	**0.160**	0.003	0.031	0.032	**0.160**	0.009	0.031	0.000
GaR_T^{VXO}	0.031	0.074	0.012	0.004	0.031	0.020	0.031	0.053	**0.160**	**0.388**
$GaR_T^{CSPREAD}$	**0.160**	0.053	**0.160**	**0.107**	**0.312**	**0.407**	**0.547**	**0.923**	**0.860**	**0.953**
GaR_T^{TERM}	**0.547**	0.001	**0.860**	0.000	**0.312**	0.003	**0.547**	0.001	**0.860**	0.014
GaR_T^{TED}	0.074	0.018	0.031	0.002	0.031	0.001	0.012	0.001	0.012	0.016
GaR_T^{ADS}	0.074	0.029	0.074	0.006	0.074	0.001	0.074	0.003	0.031	**0.109**

Note: This table shows the following two interval tests for different combined GaR: the unconditional coverage test (UC) of Kupiec (1995), which reports the probability of the null hypothesis that the exceedance is equal to the quantile (not rejecting is better); and the dynamic quantile test (DQ) of Engle and Manganelli (2004), where the null hypothesis is that the exceedance indicator is an i.i.d. process (non-rejection of the null is preferred). Bold values indicate that model passes the test with 10% probability.